



DE1 Development and Education Board

User Manual

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Chapter 1

DE1 Package

The DE1 package contains all components needed to use the DE1 board in conjunction with a computer that runs the Microsoft Windows software.

1.1 Package Contents

Figure 1.1 shows a photograph of the DE1 package.



Figure 1.1. The DE1 package contents.

The DE1 package includes:

- DE1 board
- USB Cable for FPGA programming and control
- CD-ROM containing the DE1 documentation and supporting materials, including the User Manual, the Control Panel utility, reference designs and demonstrations, device datasheets, tutorials, and a set of laboratory exercises

- CD-ROMs containing Altera's Quartus® II 6.0 Web Edition software and the Nios® II 5.0 embedded processor
- Bag of six rubber (silicon) covers for the DE1 board stands. The bag also contains some extender pins, which can be used to facilitate easier probing with testing equipment of the board's I/O expansion headers
- Clear plastic cover for the board
- 9V DC wall-mount power supply

1.2 The DE1 Board Assembly

To assemble the included stands for the DE1 board:

- Assemble a rubber (silicon) cover, as shown in Figure 1.2, for each of the six copper stands on the DE1 board
- The clear plastic cover provides extra protection, and is mounted over the top of the board by using additional stands and screws

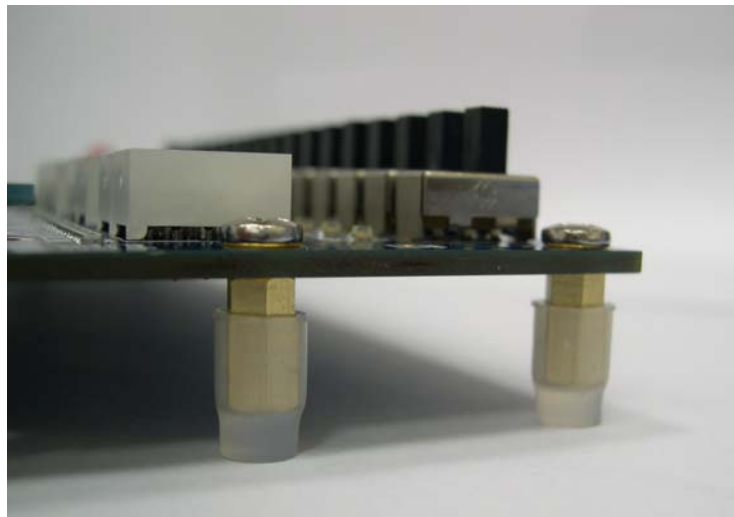


Figure 1.2. The feet for the DE1 board.

1.3 Getting Help

Here are the addresses where you can get help if you encounter problems:

- Altera Corporation
101 Innovation Drive
San Jose, California, 95134 USA
Email: university@altera.com
- Terasic Technologies
No. 356, Sec. 1, Fusing E. Rd.
Jhubei City, HsinChu County, Taiwan, 302
Email: support@terasic.com
Web: www.de1.terasic.com

Chapter 2

Altera DE1 Board

This chapter presents the features and design characteristics of the DE1 board.

2.1 Layout and Components

A photograph of the DE1 board is shown in Figure 2.1. It depicts the layout of the board and indicates the location of the connectors and key components.

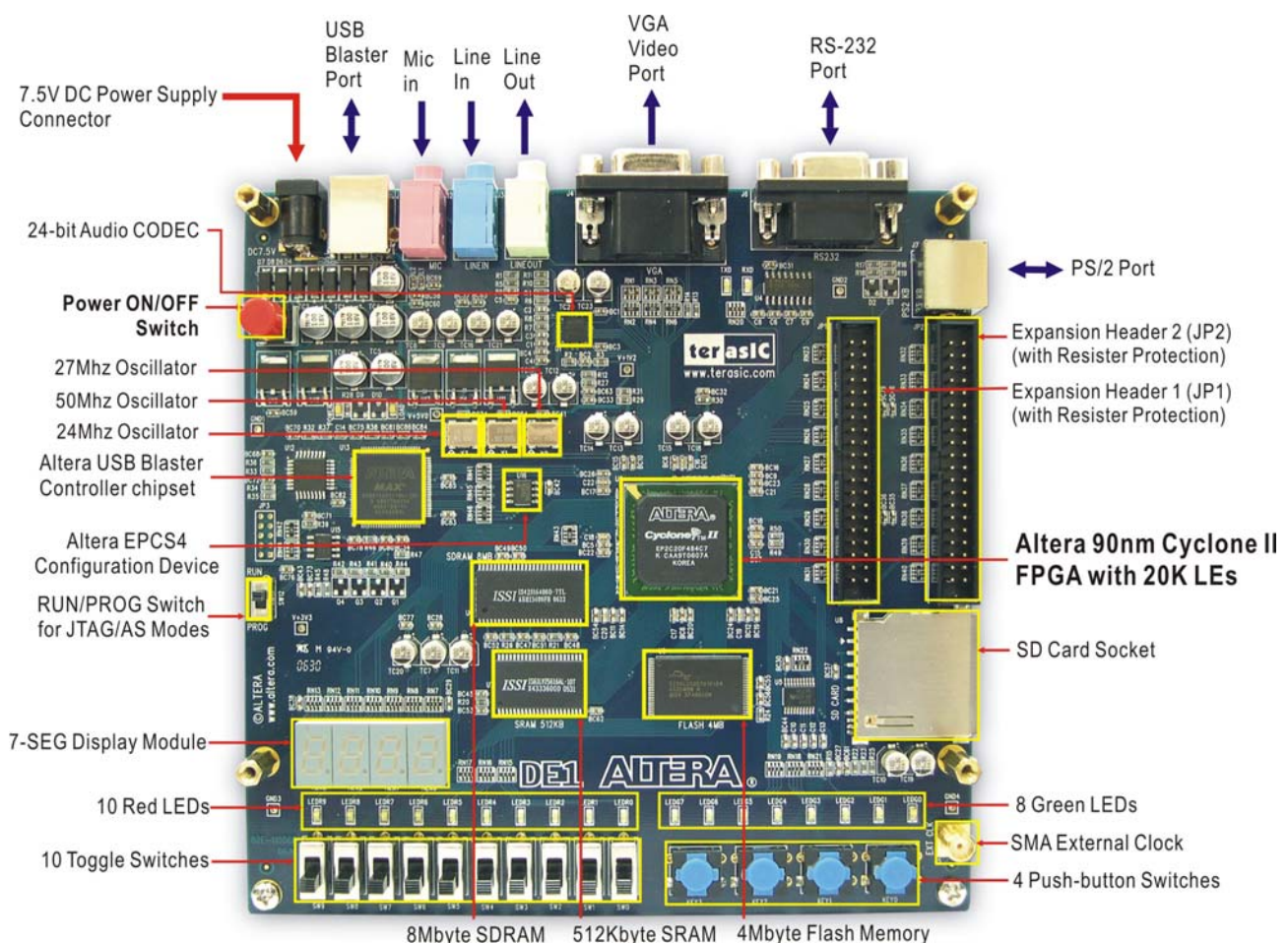


Figure 2.1. The DE1 board.

The DE1 board has many features that allow the user to implement a wide range of designed circuits, from simple circuits to various multimedia projects.

The following hardware is provided on the DE1 board:

- Altera Cyclone® II 2C20 FPGA device
- Altera Serial Configuration device – EPCS4
- USB Blaster (on board) for programming and user API control; both JTAG and Active Serial (AS) programming modes are supported
- 512-Kbyte SRAM
- 8-Mbyte SDRAM
- 4-Mbyte Flash memory
- SD Card socket
- 4 pushbutton switches
- 10 toggle switches
- 10 red user LEDs
- 8 green user LEDs
- 50-MHz oscillator, 27-MHz oscillator and 24-MHz oscillator for clock sources
- 24-bit CD-quality audio CODEC with line-in, line-out, and microphone-in jacks
- VGA DAC (4-bit resistor network) with VGA-out connector
- RS-232 transceiver and 9-pin connector
- PS/2 mouse/keyboard connector
- Two 40-pin Expansion Headers with resistor protection
- Powered by either a 7.5V DC adapter or a USB cable

In addition to these hardware features, the DE1 board has software support for standard I/O interfaces and a control panel facility for accessing various components. Also, software is provided for a number of demonstrations that illustrate the advanced capabilities of the DE1 board.

In order to use the DE1 board, the user has to be familiar with the Quartus II software. The necessary knowledge can be acquired by reading the tutorials *Getting Started with Altera's DE1 Board* and *Quartus II Introduction* (which exists in three versions based on the design entry method used, namely Verilog, VHDL or schematic entry). These tutorials are provided in the directory *DE1_tutorials* on the **DE1 System CD-ROM** that accompanies the DE1 board and can also be found on Altera's DE1 web pages.

2.2 Block Diagram of the DE1 Board

Figure 2.2 gives the block diagram of the DE1 board. To provide maximum flexibility for the user, all connections are made through the Cyclone II FPGA device. Thus, the user can configure the FPGA to implement any system design.

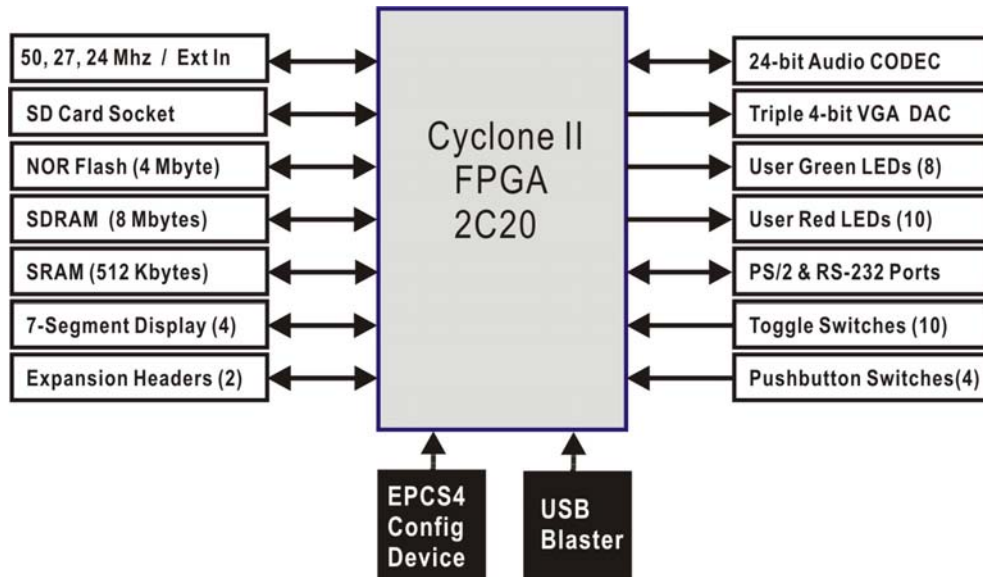


Figure 2.2. Block diagram of the DE1 board.

Following is more detailed information about the blocks in Figure 2.2:

Cyclone II 2C20 FPGA

- 18,752 LEs
- 52 M4K RAM blocks
- 240K total RAM bits
- 26 embedded multipliers
- 4 PLLs
- 315 user I/O pins
- FineLine BGA 484-pin package

Serial Configuration device and USB Blaster circuit

- Altera's EPCS4 Serial Configuration device
- On-board USB Blaster for programming and user API control
- JTAG and AS programming modes are supported

SRAM

- 512-Kbyte Static RAM memory chip
- Organized as 256K x 16 bits
- Accessible as memory for the Nios II processor and by the DE1 Control Panel

SDRAM

- 8-Mbyte Single Data Rate Synchronous Dynamic RAM memory chip
- Organized as 1M x 16 bits x 4 banks
- Accessible as memory for the Nios II processor and by the DE1 Control Panel

Flash memory

- 4-Mbyte NOR Flash memory.
- 8-bit data bus
- Accessible as memory for the Nios II processor and by the DE1 Control Panel

SD card socket

- Provides SPI mode for SD Card access
- Accessible as memory for the Nios II processor with the DE1 SD Card Driver

Pushbutton switches

- 4 pushbutton switches
- Debounced by a Schmitt trigger circuit
- Normally high; generates one active-low pulse when the switch is pressed

Toggle switches

- 10 toggle switches for user inputs
- A switch causes logic 0 when in the DOWN (closest to the edge of the DE1 board) position and logic 1 when in the UP position

Clock inputs

- 50-MHz oscillator
- 27-MHz oscillator
- 24-MHz oscillator
- SMA external clock input

Audio CODEC

- Wolfson WM8731 24-bit sigma-delta audio CODEC
- Line-level input, line-level output, and microphone input jacks
- Sampling frequency: 8 to 96 KHz

- Applications for MP3 players and recorders, PDAs, smart phones, voice recorders, etc.

VGA output

- Uses a 4-bit resistor-network DAC
- With 15-pin high-density D-sub connector
- Supports up to 640x480 at 60-Hz refresh rate
- Can be used with the Cyclone II FPGA to implement a high-performance TV Encoder

Serial ports

- One RS-232 port
- One PS/2 port
- DB-9 serial connector for the RS-232 port
- PS/2 connector for connecting a PS2 mouse or keyboard to the DE1 board

Two 40-pin expansion headers

- 72 Cyclone II I/O pins, as well as 8 power and ground lines, are brought out to two 40-pin expansion connectors
- 40-pin header is designed to accept a standard 40-pin ribbon cable used for IDE hard drives
- Resistor protection is provided

2.3 Power-up the DE1 Board

The DE1 board comes with a preloaded configuration bit stream to demonstrate some features of the board. This bit stream also allows users to see quickly if the board is working properly. To power-up the board perform the following steps:

1. Connect the provided USB cable from the host computer to the USB Blaster connector on the DE1 board. For communication between the host and the DE1 board, it is necessary to install the Altera USB Blaster driver software. If this driver is not already installed on the host computer, it can be installed as explained in the tutorial [Getting Started with Altera's DE1 Board](#). This tutorial is available on the **DE1 System CD-ROM** and from the Altera DE1 web pages.
2. Connect the 7.5V adapter to the DE1 board
3. Connect a VGA monitor to the VGA port on the DE1 board
4. Connect your headset to the Line-out audio port on the DE1 board
5. Turn the RUN/PROG switch on the left edge of the DE1 board to RUN position; the PROG position is used only for the AS Mode programming
6. Turn the power on by pressing the ON/OFF switch on the DE1 board

At this point you should observe the following:

- All user LEDs are flashing
- All 7-segment displays are cycling through the numbers 0 to F
- The VGA monitor displays the image shown in Figure 2.3 and Figure 2.4 according to SW0.
- Set the toggle switch **SW9** to the DOWN position; you should hear a 1-kHz sound
- Set the toggle switch **SW9** to the UP position and connect the output of an audio player to the Line-in connector on the DE1 board; on your headset you should hear the music played from the audio player (MP3, PC, iPod, or the like)
- You can also connect a microphone to the Microphone-in connector on the DE1 board; your voice will be mixed with the music played from the audio player



Figure 2.3. The default VGA output pattern when SW0 is set to DOWN position.



Figure 2.4. The default VGA output pattern when SW0 is set to UP position.

Chapter 3

DE1 Control Panel

The DE1 board comes with a Control Panel facility that allows a user to access various components on the board through a USB connection from a host computer. This chapter first presents some basic functions of the Control Panel, then describes its structure in block diagram form, and finally describes its capabilities.

3.1 Control Panel Setup

To run the Control Panel application, it is first necessary to configure a corresponding circuit in the Cyclone II FPGA. This is done by downloading the configuration file *DE1_USB_API.sof* into the FPGA. The downloading procedure is described in Section 4.1.

In addition to the *DE1_USB_API.sof* file, it is necessary to execute on the host computer the program *DE1_control_panel.exe*. Both of these files are available on the **DE1 System CD-ROM** that accompanies the DE1 board, in the directory *DE1_control_panel*. Of course, these files may already have been installed to some other location on your computer system.

To activate the Control Panel, perform the following steps:

1. Connect the supplied USB cable to the USB Blaster port, connect the 9V power supply, and turn the power switch ON
2. Set the RUN/PROG switch to the RUN position
3. Start the Quartus II software
4. Select **Tools > Programmer** to reach the window in Figure 3.1. Click on **Add File** and in the pop-up window that appears select the *DE1_USB_API.sof* file. Next, click on the **Program/Configure** box which results in the image displayed in the figure. Now, click **Start** to download the configuration file into the FPGA.
5. Start the executable *DE1_control_panel.exe* on the host computer. The Control Panel user interface shown in Figure 3.2 will appear.
6. Open the USB port by clicking **Open > Open USB Port 0**. The DE1 Control Panel application will list all the USB ports that connect to DE1 boards. The DE1 Control Panel can control up to 4 DE1 boards using the USB links. **The Control Panel will occupy the USB port until you close that port; you cannot use Quartus II to download a configuration file into the FPGA until you close the USB port.**

7. The Control Panel is now ready for use; experiment by setting the value of some 7-segment display and observing the result on the DE1 board.

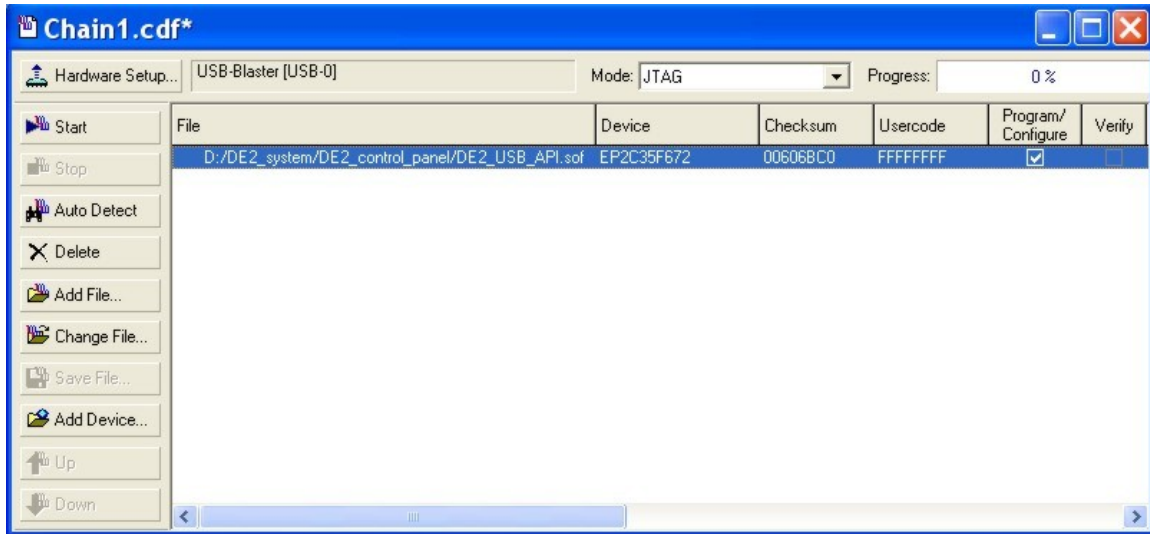


Figure 3.1. Quartus II Programmer window.

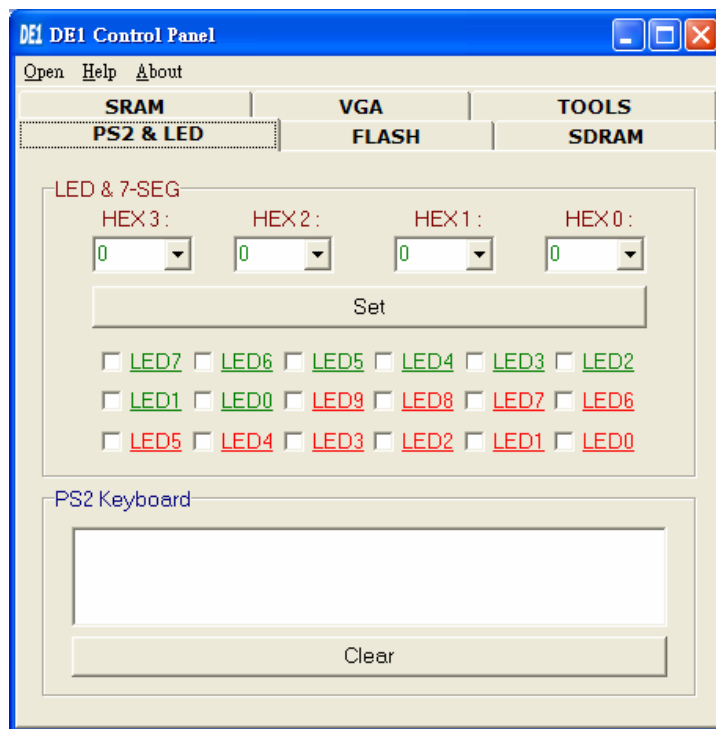


Figure 3.2. The DE1 Control Panel.

The concept of the DE1 Control Panel is illustrated in Figure 3.3. The IP that performs the control functions is implemented in the FPGA device. It communicates with the Control Panel window, which is active on the host computer, via the USB Blaster link. The graphical interface is used to

issue commands to the control circuitry. The provided IP handles all requests and performs data transfers between the computer and the DE1 board.

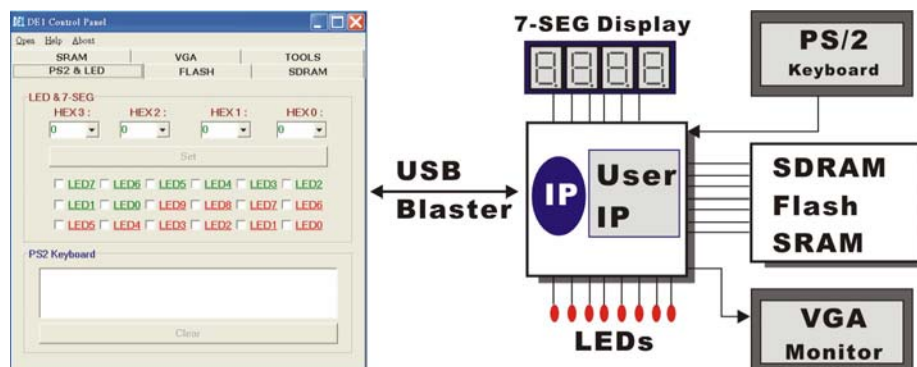


Figure 3.3. The DE1 Control Panel concept.

The DE1 Control Panel can be used to change the values displayed on 7-segment displays, light up LEDs, talk to the PS/2 keyboard, read/write the SRAM, Flash Memory and SDRAM, load an image pattern to display as VGA output, load music to the memory and play music via the audio DAC. The feature of reading/writing a byte or an entire file from/to the Flash Memory allows the user to develop multimedia applications (Flash Audio Player, Flash Picture Viewer) without worrying about how to build a Flash Memory Programmer.

3.2 Controlling the LEDs and 7-Segment Displays

A simple function of the Control Panel is to allow setting the values displayed on LEDs and 7-segment displays.

In the window shown in Figure 3.2, the values to be displayed by the 7-segment displays (which are named *HEX7-0*) can be entered into the corresponding boxes and displayed by pressing the **Set** button.

Choosing the **LED** tab leads to the window in Figure 3.4. Here, you can turn the individual LEDs on by selecting them and pressing the **Set** button.

The ability to set arbitrary values into simple display devices is not needed in typical design activities. However, it gives the user a simple mechanism for verifying that these devices are functioning correctly in case a malfunction is suspected. Thus, it can be used for troubleshooting purposes.

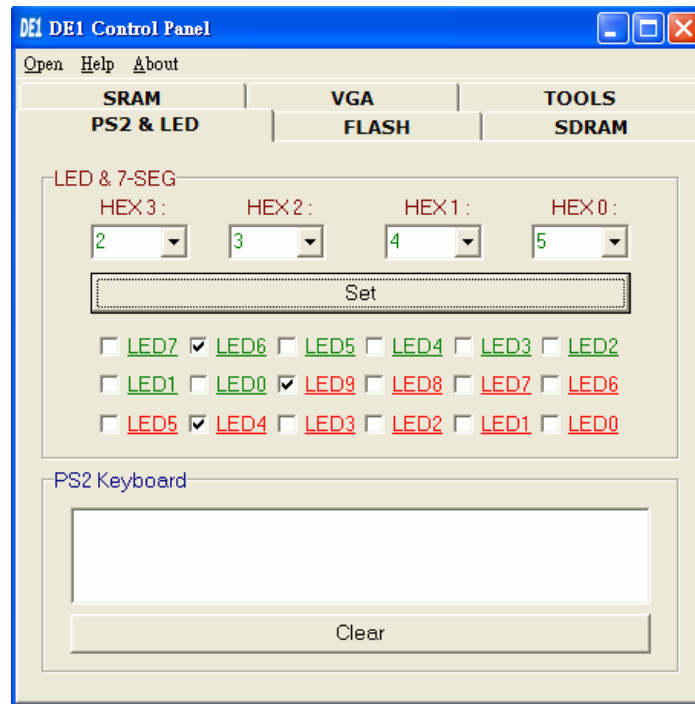


Figure 3.4. Controlling LEDs and the LCD display.

3.3 SDRAM/SRAM Controller and Programmer

The Control Panel can be used to write/read data to/from the SDRAM and SRAM chips on the DE1 board. We will describe how the SDRAM may be accessed; the same approach is used to access the SRAM. Click on the SDRAM tab to reach the window in Figure 3.5.

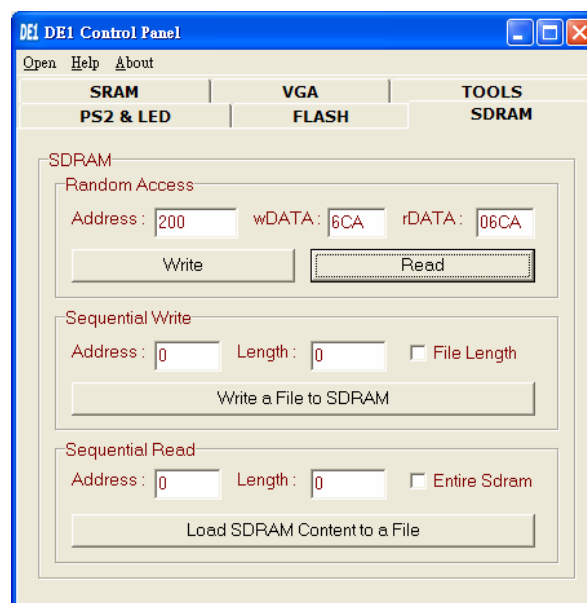


Figure 3.5. Accessing the SDRAM.

A 16-bit word can be written into the SDRAM by entering the address of the desired location, specifying the data to be written, and pressing the **Write** button. Contents of the location can be read by pressing the **Read** button. Figure 3.5 depicts the result of writing the hexadecimal value 6CA into location 200, followed by reading the same location.

The Sequential Write function of the Control Panel is used to write the contents of a file into the SDRAM as follows:

1. Specify the starting address in the **Address** box.
2. Specify the number of bytes to be written in the **Length** box. If the entire file is to be loaded, then a checkmark may be placed in the **File Length** box instead of giving the number of bytes.
3. To initiate the writing of data, click on the **Write a File to SDRAM** button.
4. When the Control Panel responds with the standard Windows dialog box asking for the source file, specify the desired file in the usual manner.

The Control Panel also supports loading files with a *.hex* extension. Files with a *.hex* extension are ASCII text files that specify memory values using ASCII characters to represent hexadecimal values. For example, a file containing the line

0123456789ABCDEF

defines four 16-bit values: 0123, 4567, 89AB, CDEF. These values will be loaded consecutively into the memory.

The Sequential Read function is used to read the contents of the SDRAM and place them into a file as follows:

1. Specify the starting address in the **Address** box.
2. Specify the number of bytes to be copied into the file in the **Length** box. If the entire contents of the SDRAM are to be copied (which involves all 8 Mbytes), then place a checkmark in the **Entire SDRAM** box.
3. Press **Load SDRAM Content to a File** button.
4. When the Control Panel responds with the standard Windows dialog box asking for the destination file, specify the desired file in the usual manner.

3.4 Flash Memory Programmer

The Control Panel can be used to write/read data to/from the Flash memory chip on the DE1 board. It can be used to:

- Erase the entire Flash memory
- Write one byte to the memory

- Read one byte from the memory
- Write a binary file to the memory
- Load the contents of the Flash memory into a file

Note the following characteristics of the Flash memory:

- The Flash memory chip is organized as 4 M x 8 bits.
- You must erase the entire Flash memory before you can write into it. (Be aware that the number of times a Flash memory can be erased is limited.)
- The time required to erase the entire Flash memory is about 20 seconds. Do not close the DE1 Control Panel in the middle of the operation.

To open the Flash memory control window, shown in Figure 3.6, select the FLASH tab in the Control Panel.

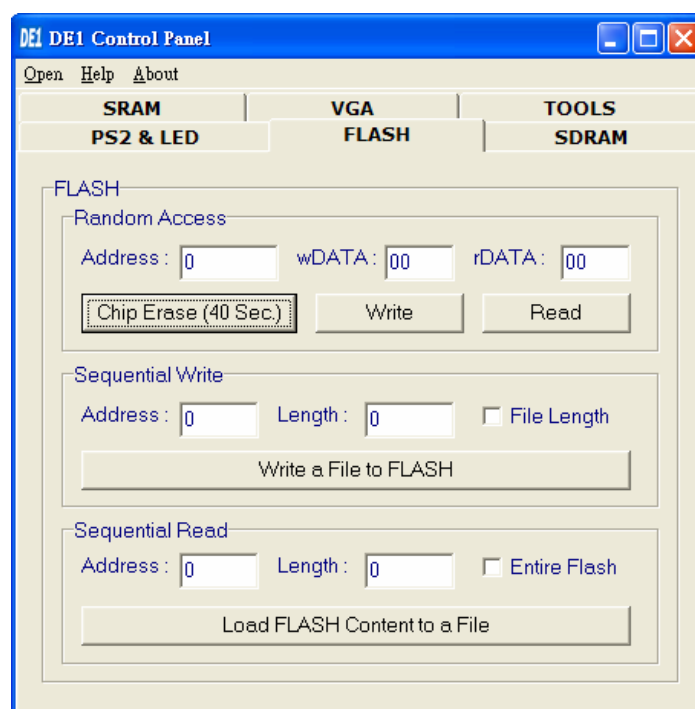


Figure 3.6. Flash memory control window.

A byte of data can be written into a random location on the Flash chip as follows:

1. Click on the **Chip Erase** button. The button and the window frame title will prompt you to wait until the operation is finished, which takes about 20 seconds.
2. Enter the desired address into the **Address** box and the data byte into the **wDATA** box. Then, click on the **Write** button.

To read a byte of data from a random location, enter the address of the location and click on the **Read** button. The **rDATA** box will display the data read back from the address specified.

The Sequential Write function is used to load a file into the Flash chip as follows:

1. Specify the starting address and the length of data (in bytes) to be written into the Flash memory. You can click on the **File Length** checkbox to indicate that you want to load the entire file.
2. Click on the **Write a File to Flash** button to activate the writing process.
3. When the Control Panel responds with the standard Windows dialog box asking for the source file, specify the desired file in the usual manner.

The Sequential Read function is used to read the data stored in the Flash memory and write this data into a file as follows:

1. Specify the starting address and the length of data (in bytes) to be read from the Flash memory. You can click on the **Entire Flash** checkbox to indicate that you want to copy the entire contents of the Flash memory into a specified file.
2. Click on the **Load Flash Content to a File** button to activate the reading process.
3. When the Control Panel responds with the standard Windows dialog box asking for the destination file, specify the desired file in the usual manner.

3.5 Overall Structure of the DE1 Control Panel

The DE1 Control Panel facility communicates with a circuit that is instantiated in the Cyclone II FPGA. This circuit is specified in Verilog code, which makes it possible for a knowledgeable user to change the functionality of the Control Panel. The code is located inside the *DE1_demonstrations* directory on the **DE1 System CD-ROM**.

To run the Control Panel, the user must first set it up as explained in Section 3.1. Figure 3.7 depicts the structure of the Control Panel. Each input/output device is controlled by a controller instantiated in the FPGA chip. The communication with the PC is done via the USB Blaster link. A Command Controller circuit interprets the commands received from the PC and performs the appropriate actions. The SDRAM, SRAM, and Flash Memory controllers have three user-selectable asynchronous ports in addition to the Host port that provides a link with the Command Controller. The connection between the VGA DAC Controller and the FPGA memory allows displaying of the default image shown on the left side of the figure, which is stored in an M4K block in the Cyclone II chip. The connection between the Audio DAC Controller and a lookup table in the FPGA is used to produce a test audio signal of 1 kHz.

To let users implement and test their IP cores (written in Verilog) without requiring them to implement complex API/Host control software and memory (SRAM/SDRAM/Flash) controllers, we provide an integrated control environment consisting of a software controller in C++, a USB command controller, and a multi-port SRAM/SDRAM/Flash controller.

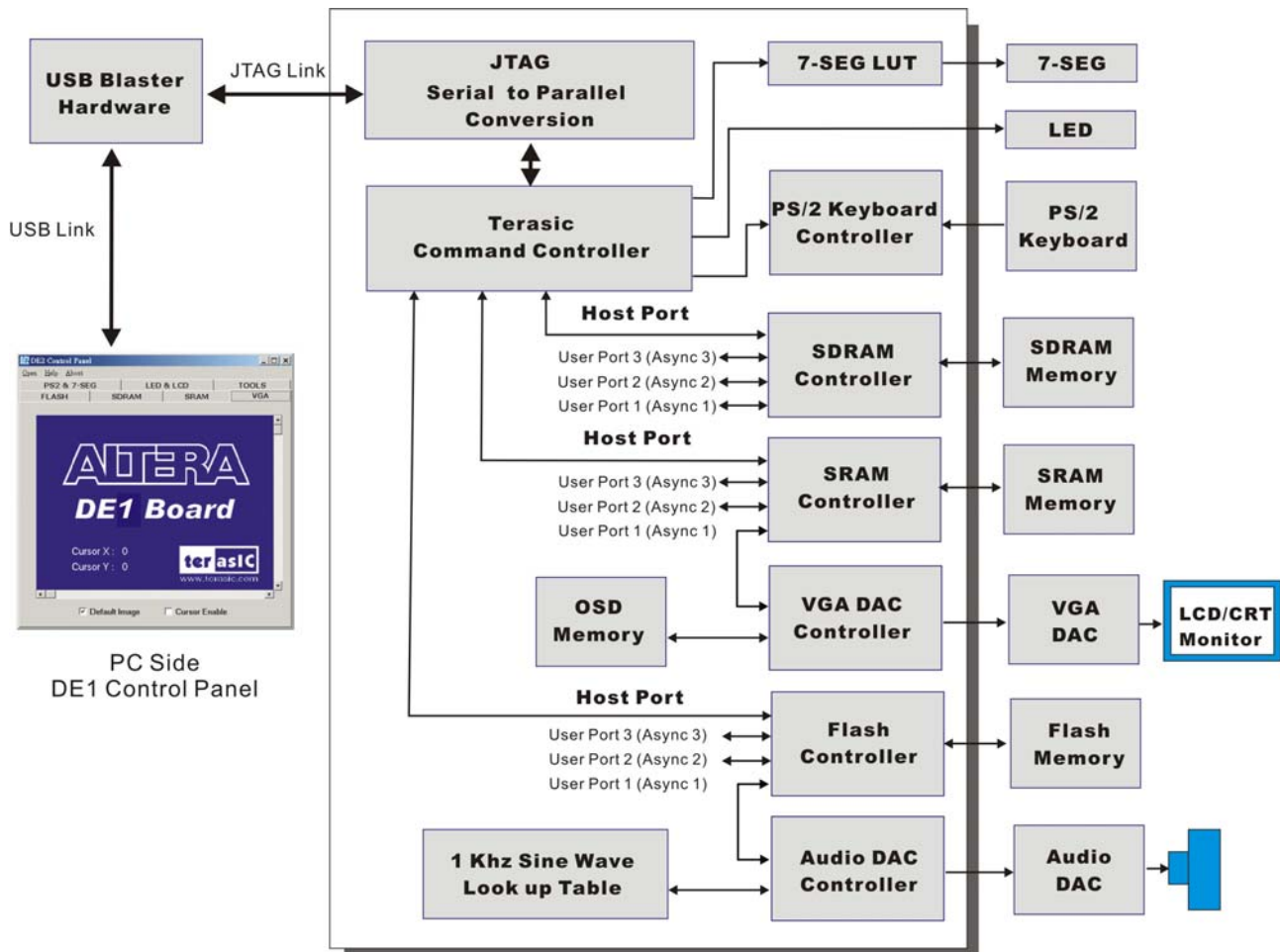


Figure 3.7. The DE1 Control Panel block diagram.

Users can connect circuits of their own design to one of the User Ports of the SRAM/SDRAM/Flash controller. Then, they can download binary data into the SRAM/SDRAM/Flash. Once the data is downloaded to the SDRAM/Flash, users can configure the memory controllers so that their circuits can read/write the SDRAM/Flash via the User Ports connected.

3.6 TOOLS – Multi-Port SRAM/SDRAM/Flash Controller

The TOOLS page of the Control Panel GUI allows selection of the User Ports. We will illustrate a typical process by implementing a Flash Music Player. The music data is loaded into the Flash memory. User Port 1 in the Flash Controller is used to send the music data to the Audio DAC Controller and hence to the audio output jack.

You can implement this application as follows:

1. Erase the Flash memory (as explained in Section 3.4). Then, write a music file into the Flash memory. You can use the file *music.wav* in the directory *DE1_demonstrations\music* on the **DE1 System CD-ROM**.
2. In the DE1 Control Panel, select the TOOLS tab to reach the window in Figure 3.8.

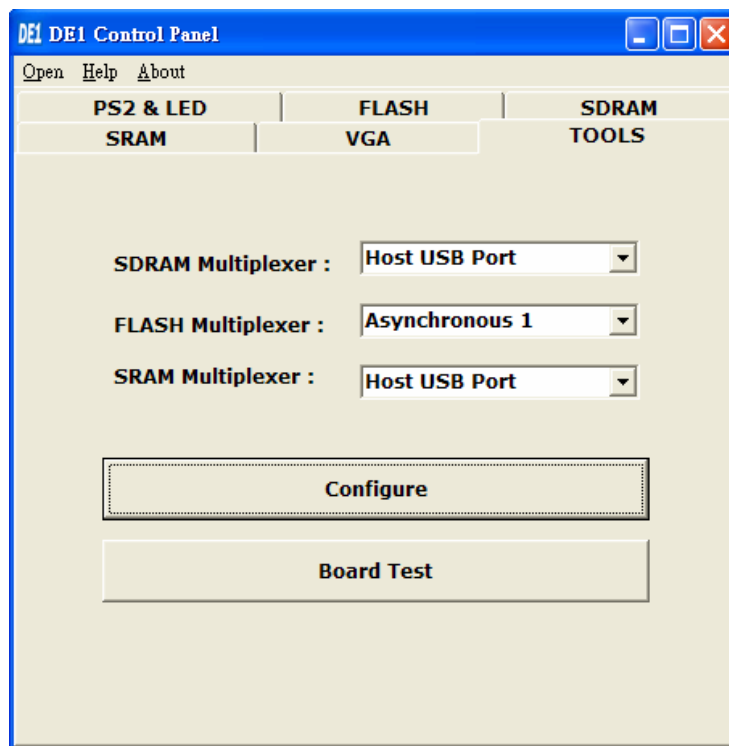


Figure 3.8. TOOLS window of the DE1 Control Panel.

3. Select the Asynchronous 1 port for the Flash Multiplexer and then click on the **Configure** button to activate the port. You need to click the Configure button to enable the connection from the Flash Memory to the Asynchronous Port 1 of the Flash Controller (indicated in Figure 3.7).
4. Set toggle switches SW1 and SW0 to OFF (DOWN position) and ON (UP position), respectively.
5. Plug your headset or a speaker into the audio output jack and you should hear the music

played from the Audio DAC circuit.

6. Note that the Asynchronous Port 1 is connected to the Audio DAC part, as shown in Figure 3.7. Once you selected Asynchronous Port 1 and clicked the Configure button, the Audio DAC Controller will communicate with the Flash memory directly. In our example, the *AUDIO_DAC* Verilog module defines a circuit that reads the contents of the Flash memory and sends it to the external audio chip.

3.7 VGA Display Control

The Control Panel provides a tool with the associated IP that allows the user to display an image via the VGA output port. To illustrate this feature, we will show how an image can be displayed on a VGA monitor. Perform the following steps to display a default image:

- Select the VGA tab in the Control Panel to reach the window in Figure 3.9.



Figure 3.9. Displayed image and the cursor controlled by the scroll bars

- Make sure that the checkboxes **Default Image** and **Cursor Enable** are checked.
- Connect a VGA monitor to the DE1 board and you should see on the screen the default image shown in Figure 3.9. The image includes a cursor which can be controlled by means of the X/Y-axes scroll bars on the DE1 Control Panel.

The image in Figure 3.9 is stored in an M4K memory block in the Cyclone II FPGA. It is loaded into the M4K block in the MIF/Hex(Intel) format during the default bit stream configuration stage. We will next describe how you can display other images and use your own images to generate the binary data patterns that can be displayed on the VGA monitor.

Another image is provided in the file *picture.dat* in the folder *DE1_demonstrations\pictures* on the **DE1 System CD-ROM**. You can display this image as follows:

- Select the **SRAM** page of the Control Panel and load the file *picture.dat* into the SRAM.
- Select the **TOOLS** page and choose **Asynchronous 1** for the **SRAM multiplexer** port as shown in Figure 3.10. Click on the **Configure** button to activate the multi-port setup.

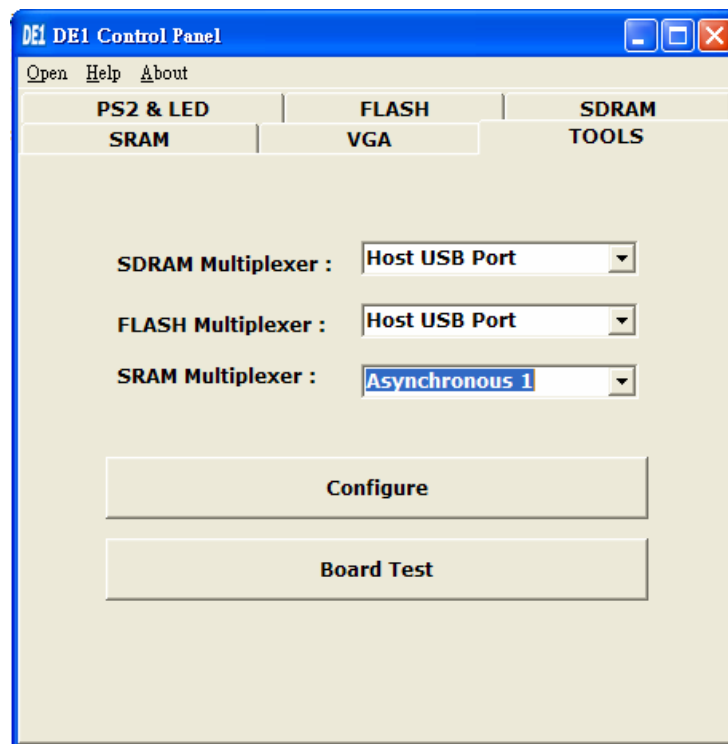


Figure 3.10. Use the Asynchronous Port 1 to access the image data in the SRAM.

- The FPGA is now configured as indicated in Figure 3.11.
- Select the **VGA** page and deselect the checkbox **Default Image**.
- The VGA monitor should display the *picture.dat* image from the SRAM, as depicted in Figure 3.12. You can turn off the cursor by deselecting the **Cursor Enable** checkbox.

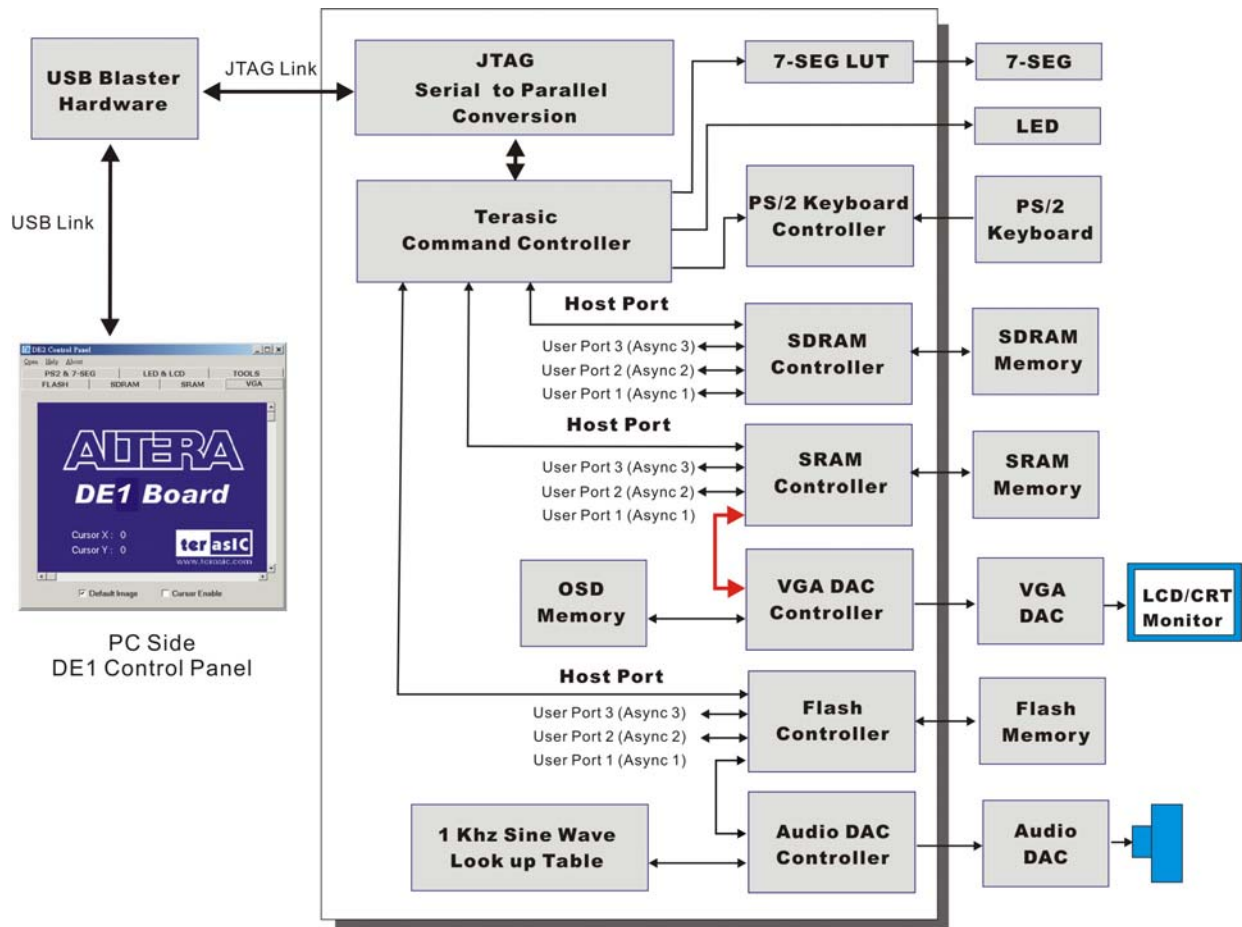


Figure 3.11. Multi-Port Controller configured to display an image from the SRAM.



Figure 3.12. A displayed image.

You can display any image file by loading it into the SRAM chip or into an M4K memory block in the Cyclone II chip. This requires generating a bitmap file, which may be done as follows:

1. Load the desired image into an image processing tool, such as Corel PhotoPaint.
2. Resample the original image to have a 640 x 480 resolution. Save the modified image in the Windows Bitmap format.
3. Execute *DE1_control_panel\ImgConv.exe*, an image conversion tool developed for the DE1 board, to reach the window in Figure 3.13.
4. Click on the **Open Bitmap** button and select the 640 x 480 Grayscale photo for conversion.
5. When the processing of the file is completed, click on the **Save Raw Data** button and a file named *Raw_Data_Gray.dat* will be generated and stored in the same directory as the original image file. You can change the file name prefix from *Raw_Data* to another name by changing the File Name field in the displayed window.
6. *Raw_Data_Gray.dat* is the raw data that can be downloaded directly into the SRAM on the DE1 board and displayed on the VGA monitor using the VGA controller IP described in the *DE1_USB_API* project.
7. The *ImgConv* tool will also generate *Raw_Data_BW.dat* (and its corresponding TXT format) for the black and white version of the image – the threshold for judging black or white level is defined in the BW Threshold.

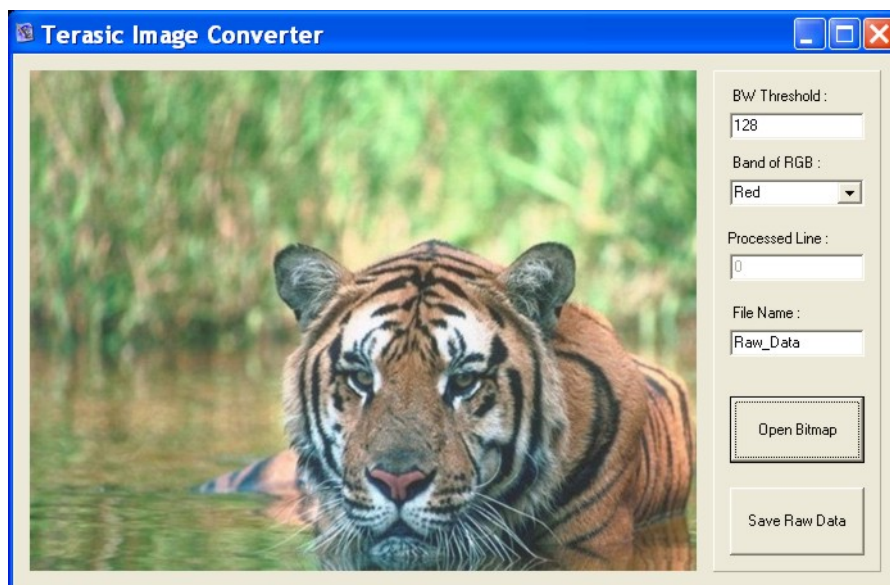


Figure 3.13. The image converter window.

Image Source	R/G/B Band Filter	B&W Threshold Filter	Output Result (640x480)
Color Picture	R/G/B	N/A	Raw_Data_Gray
Color Picture	R/G/B (optional)	BW Threshold	Raw_Data_BW + Raw_Data_BW.txt
Grayscale Picture	N/A	N/A	Raw_Data_Gray
Grayscale Picture	N/A	BW Threshold	Raw_Data_BW + Raw_Data_BW.txt

Note: Raw_Data_BW.txt is used to fill in the MIF/Intel Hex format for M4K SRAM

Chapter 4

Using the DE1 Board

This chapter gives instructions for using the DE1 board and describes each of its I/O devices.

4.1 Configuring the Cyclone II FPGA

The procedure for downloading a circuit from a host computer to the DE1 board is described in the tutorial *Quartus II Introduction*. This tutorial is found in the *DE1_tutorials* folder on the **DE1 System CD-ROM**, and it is also available on the Altera DE1 web pages. The user is encouraged to read the tutorial first, and to treat the information below as a short reference.

The DE1 board contains a serial EEPROM chip that stores configuration data for the Cyclone II FPGA. This configuration data is automatically loaded from the EEPROM chip into the FPGA each time power is applied to the board. Using the Quartus II software, it is possible to reprogram the FPGA at any time, and it is also possible to change the non-volatile data that is stored in the serial EEPROM chip. Both types of programming methods are described below.

1. *JTAG* programming: In this method of programming, named after the IEEE standards *Joint Test Action Group*, the configuration bit stream is downloaded directly into the Cyclone II FPGA. The FPGA will retain this configuration as long as power is applied to the board; the configuration is lost when the power is turned off.
2. *AS* programming: In this method, called *Active Serial* programming, the configuration bit stream is downloaded into the Altera EPCS4 serial EEPROM chip. It provides non-volatile storage of the bit stream, so that the information is retained even when the power supply to the DE1 board is turned off. When the board's power is turned on, the configuration data in the EPCS4 device is automatically loaded into the Cyclone II FPGA.

The sections below describe the steps used to perform both JTAG and AS programming. For both methods the DE1 board is connected to a host computer via a USB cable. Using this connection, the board will be identified by the host computer as an Altera *USB Blaster* device. The process for installing on the host computer the necessary software device driver that communicates with the USB Blaster is described in the tutorial *Getting Started with Altera's DE1 Board*. This tutorial is available on the **DE1 System CD-ROM** and from the Altera DE1 web pages.

Configuring the FPGA in JTAG Mode

Figure 4.1 illustrates the JTAG configuration setup. To download a configuration bit stream into the Cyclone II FPGA, perform the following steps:

- Ensure that power is applied to the DE1 board
- Connect the supplied USB cable to the USB Blaster port on the DE1 board (see Figure 2.1)
- Configure the JTAG programming circuit by setting the RUN/PROG switch (on the left side of the board) to the RUN position.
- The FPGA can now be programmed by using the Quartus II Programmer module to select a configuration bit stream file with the *.sof* filename extension

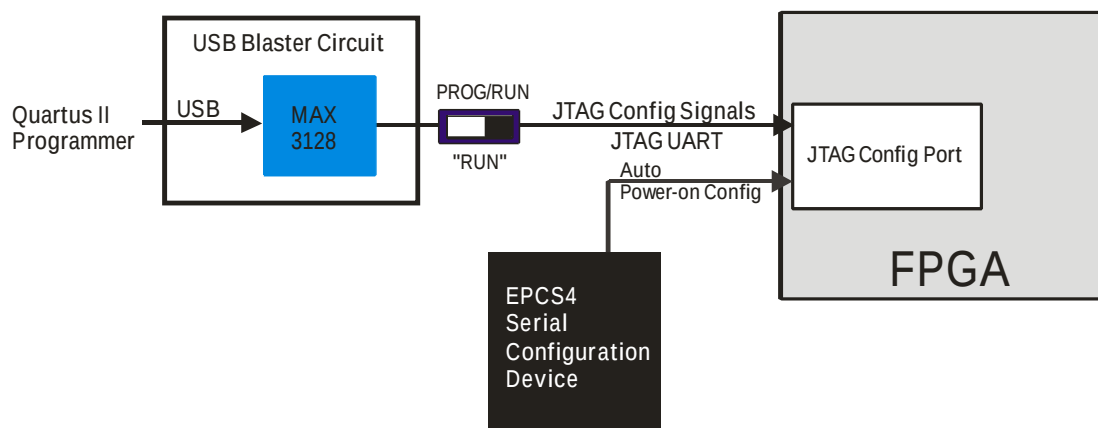


Figure 4.1. The JTAG configuration scheme.

Configuring the EPCS4 in AS Mode

Figure 4.2 illustrates the AS configuration set up. To download a configuration bit stream into the EPCS4 serial EEPROM device, perform the following steps:

- Ensure that power is applied to the DE1 board
- Connect the supplied USB cable to the USB Blaster port on the DE1 board (see Figure 2.1)
- Configure the JTAG programming circuit by setting the RUN/PROG switch (on the left side of the board) to the PROG position.
- The EPCS4 chip can now be programmed by using the Quartus II Programmer module to select a configuration bit stream file with the *.pof* filename extension
- Once the programming operation is finished, set the RUN/PROG switch back to the RUN position and then reset the board by turning the power switch off and back on; this action causes the new configuration data in the EPCS4 device to be loaded into the FPGA chip.

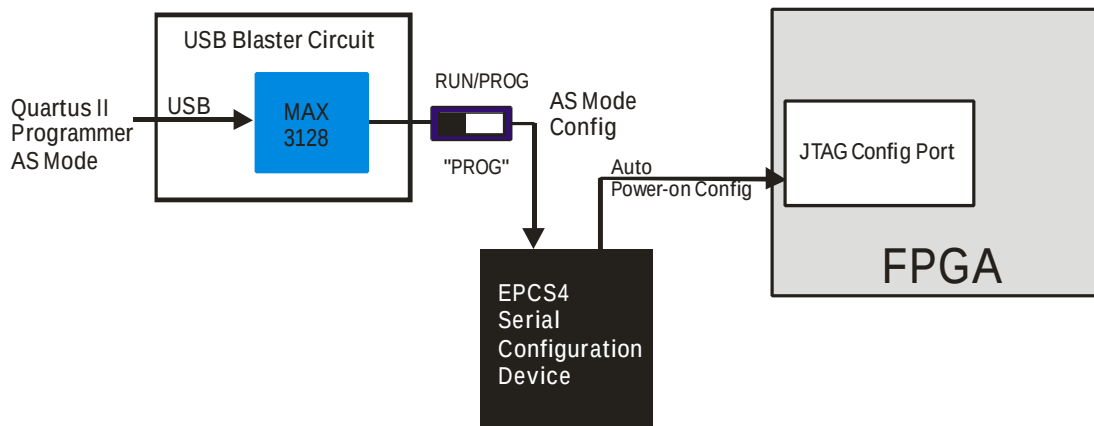


Figure 4.2. The AS configuration scheme.

In addition to its use for JTAG and AS programming, the USB Blaster port on the DE1 board can also be used to control some of the board's features remotely from a host computer. Details that describe this method of using the USB Blaster port are given in Chapter 3.

4.2 Using the LEDs and Switches

The DE1 board provides four pushbutton switches. Each of these switches is debounced using a Schmitt Trigger circuit, as indicated in Figure 4.3. The four outputs called *KEY0*, ..., *KEY3* of the Schmitt Trigger device are connected directly to the Cyclone II FPGA. Each switch provides a high logic level (3.3 volts) when it is not pressed, and provides a low logic level (0 volts) when depressed. Since the pushbutton switches are debounced, they are appropriate for use as clock or reset inputs in a circuit.

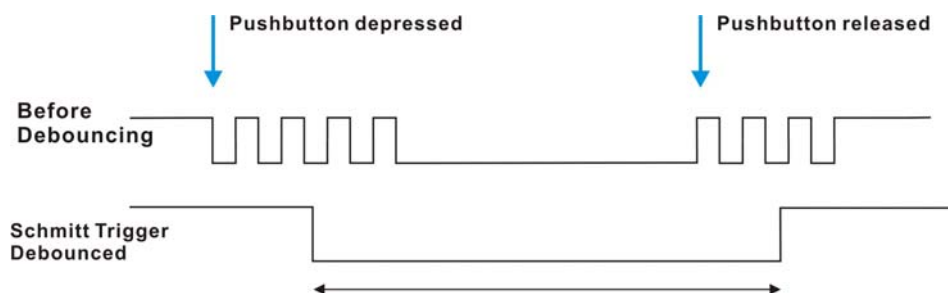


Figure 4.3. Switch debouncing.

There are also 10 toggle switches (sliders) on the DE1 board. These switches are not debounced, and are intended for use as level-sensitive data inputs to a circuit. Each switch is connected directly to a pin on the Cyclone II FPGA. When a switch is in the DOWN position (closest to the edge of the board) it provides a low logic level (0 volts) to the FPGA, and when the switch is in the UP position it provides a high logic level (3.3 volts).

There are 27 user-controllable LEDs on the DE1 board. Eighteen red LEDs are situated above the 18 toggle switches, and eight green LEDs are found above the pushbutton switches (the 9th green LED is in the middle of the 7-segment displays). Each LED is driven directly by a pin on the Cyclone II FPGA; driving its associated pin to a high logic level turns the LED on, and driving the pin low turns it off. A schematic diagram that shows the pushbutton and toggle switches is given in Figure 4.4. A schematic diagram that shows the LED circuitry appears in Figure 4.5.

A list of the pin names on the Cyclone II FPGA that are connected to the toggle switches is given in Table 4.1. Similarly, the pins used to connect to the pushbutton switches and LEDs are displayed in Tables 4.2 and 4.3, respectively.

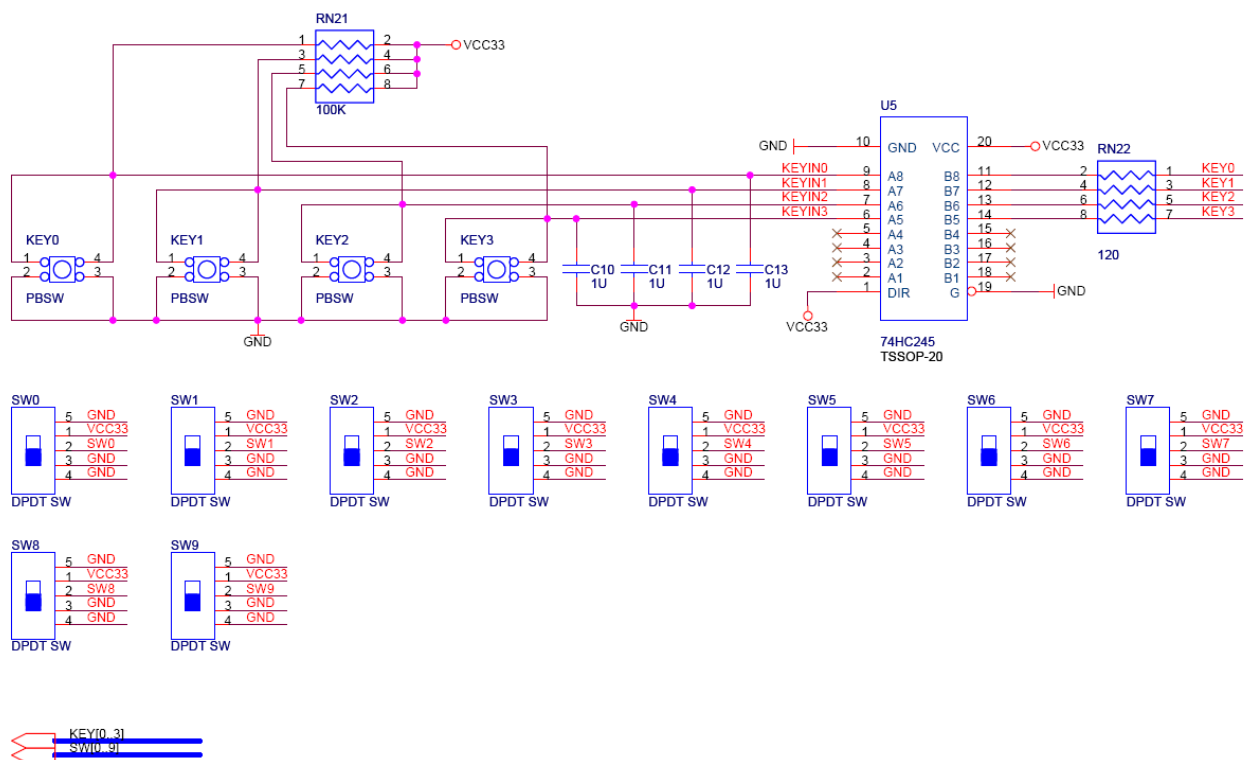


Figure 4.4. Schematic diagram of the pushbutton and toggle switches.

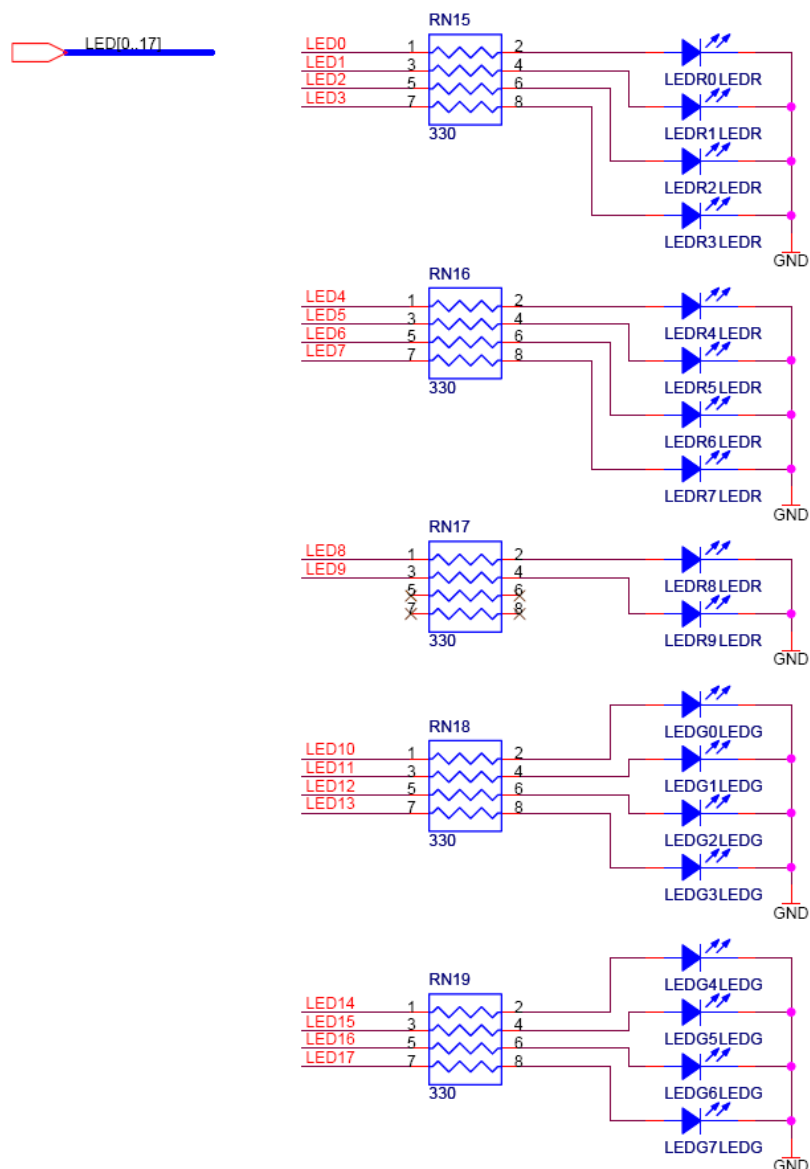


Figure 4.5. Schematic diagram of the LEDs.

Signal Name	FPGA Pin No.	Description
SW[0]	PIN_L22	Toggle Switch[0]
SW[1]	PIN_L21	Toggle Switch[1]
SW[2]	PIN_M22	Toggle Switch[2]
SW[3]	PIN_V12	Toggle Switch[3]
SW[4]	PIN_W12	Toggle Switch[4]
SW[5]	PIN_U12	Toggle Switch[5]
SW[6]	PIN_U11	Toggle Switch[6]
SW[7]	PIN_M2	Toggle Switch[7]

SW[8]	PIN_M1	Toggle Switch[8]
SW[9]	PIN_L2	Toggle Switch[9]

Table 4.1. Pin assignments for the toggle switches.

Signal Name	FPGA Pin No.	Description
KEY[0]	PIN_R22	Pushbutton[0]
KEY[1]	PIN_R21	Pushbutton[1]
KEY[2]	PIN_T22	Pushbutton[2]
KEY[3]	PIN_T21	Pushbutton[3]

Table 4.2. Pin assignments for the pushbutton switches.

Signal Name	FPGA Pin No.	Description
LEDR[0]	PIN_R20	LED Red[0]
LEDR[1]	PIN_R19	LED Red[1]
LEDR[2]	PIN_U19	LED Red[2]
LEDR[3]	PIN_Y19	LED Red[3]
LEDR[4]	PIN_T18	LED Red[4]
LEDR[5]	PIN_V19	LED Red[5]
LEDR[6]	PIN_Y18	LED Red[6]
LEDR[7]	PIN_U18	LED Red[7]
LEDR[8]	PIN_R18	LED Red[8]
LEDR[9]	PIN_R17	LED Red[9]
LEDG[0]	PIN_U22	LED Green[0]
LEDG[1]	PIN_U21	LED Green[1]
LEDG[2]	PIN_V22	LED Green[2]
LEDG[3]	PIN_V21	LED Green[3]
LEDG[4]	PIN_W22	LED Green[4]
LEDG[5]	PIN_W21	LED Green[5]
LEDG[6]	PIN_Y22	LED Green[6]
LEDG[7]	PIN_Y21	LED Green[7]

Table 4.3. Pin assignments for the LEDs.

4.3 Using the 7-segment Displays

The DE1 Board has four 7-segment displays. These displays are arranged into a group of four, with the intent of displaying numbers of various sizes. As indicated in the schematic in Figure 4.6, the seven segments are connected to pins on the Cyclone II FPGA. Applying a low logic level to a segment causes it to light up, and applying a high logic level turns it off.

Each segment in a display is identified by an index from 0 to 6, with the positions given in Figure 4.7. Note that the dot in each display is unconnected and cannot be used. Table 4.4 shows the assignments of FPGA pins to the 7-segment displays.

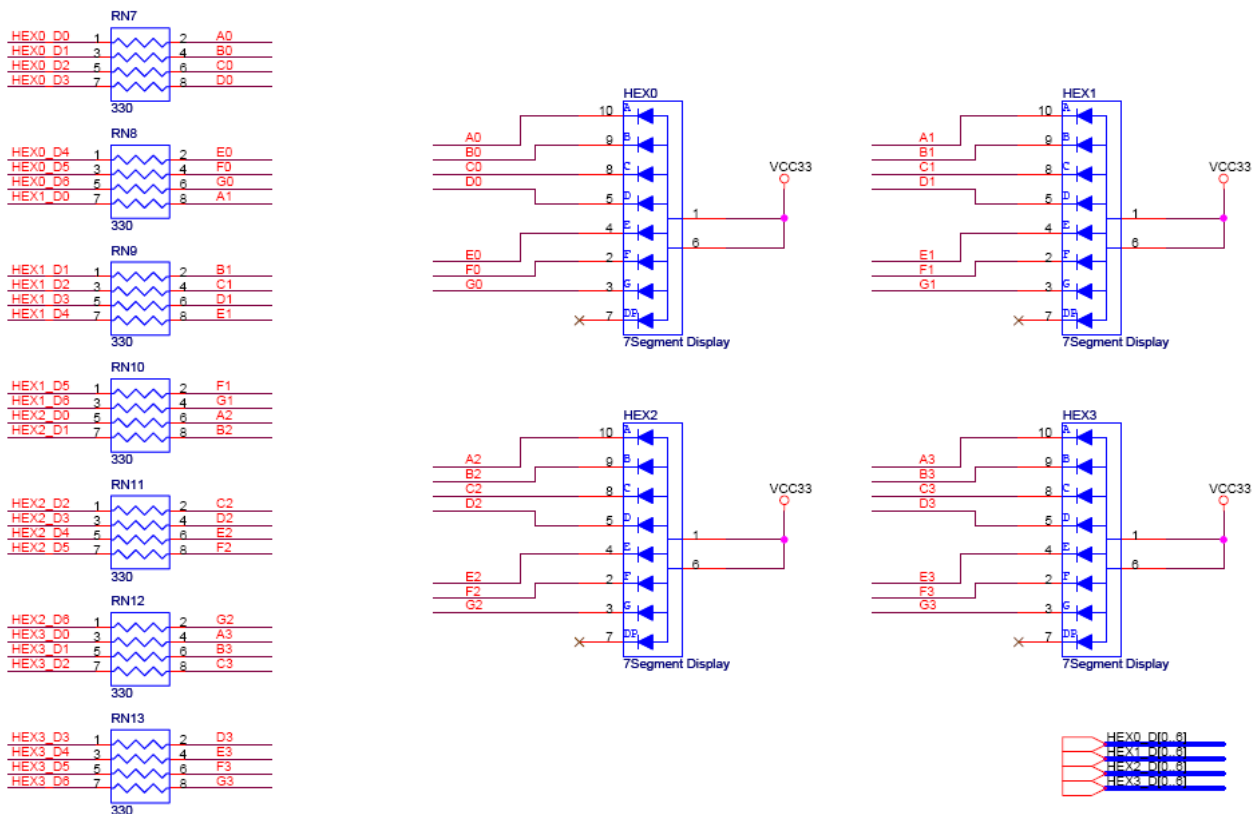


Figure 4.6. Schematic diagram of the 7-segment displays.

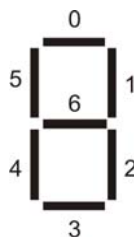


Figure 4.7. Position and index of each segment in a 7-segment display.

Signal Name	FPGA Pin No.	Description
HEX0[0]	PIN_J2	Seven Segment Digit 0[0]
HEX0[1]	PIN_J1	Seven Segment Digit 0[1]
HEX0[2]	PIN_H2	Seven Segment Digit 0[2]
HEX0[3]	PIN_H1	Seven Segment Digit 0[3]
HEX0[4]	PIN_F2	Seven Segment Digit 0[4]
HEX0[5]	PIN_F1	Seven Segment Digit 0[5]
HEX0[6]	PIN_E2	Seven Segment Digit 0[6]
HEX1[0]	PIN_E1	Seven Segment Digit 1[0]
HEX1[1]	PIN_H6	Seven Segment Digit 1[1]
HEX1[2]	PIN_H5	Seven Segment Digit 1[2]
HEX1[3]	PIN_H4	Seven Segment Digit 1[3]
HEX1[4]	PIN_G3	Seven Segment Digit 1[4]
HEX1[5]	PIN_D2	Seven Segment Digit 1[5]
HEX1[6]	PIN_D1	Seven Segment Digit 1[6]
HEX2[0]	PIN_G5	Seven Segment Digit 2[0]
HEX2[1]	PIN_G6	Seven Segment Digit 2[1]
HEX2[2]	PIN_C2	Seven Segment Digit 2[2]
HEX2[3]	PIN_C1	Seven Segment Digit 2[3]
HEX2[4]	PIN_E3	Seven Segment Digit 2[4]
HEX2[5]	PIN_E4	Seven Segment Digit 2[5]
HEX2[6]	PIN_D3	Seven Segment Digit 2[6]
HEX3[0]	PIN_F4	Seven Segment Digit 3[0]
HEX3[1]	PIN_D5	Seven Segment Digit 3[1]
HEX3[2]	PIN_D6	Seven Segment Digit 3[2]
HEX3[3]	PIN_J4	Seven Segment Digit 3[3]
HEX3[4]	PIN_L8	Seven Segment Digit 3[4]
HEX3[5]	PIN_F3	Seven Segment Digit 3[5]
HEX3[6]	PIN_D4	Seven Segment Digit 3[6]

Table 4.4. Pin assignments for the 7-segment displays.

4.4 Clock Inputs

The DE1 board includes **three oscillators that produce 27 MHz, 24Mhz, and 50 MHz clock signals**. The board also includes an SMA connector which can be used to connect an external clock source to the board. The schematic of the clock circuitry is shown in Figure 4.8, and the associated pin assignments appear in Table 4.5.

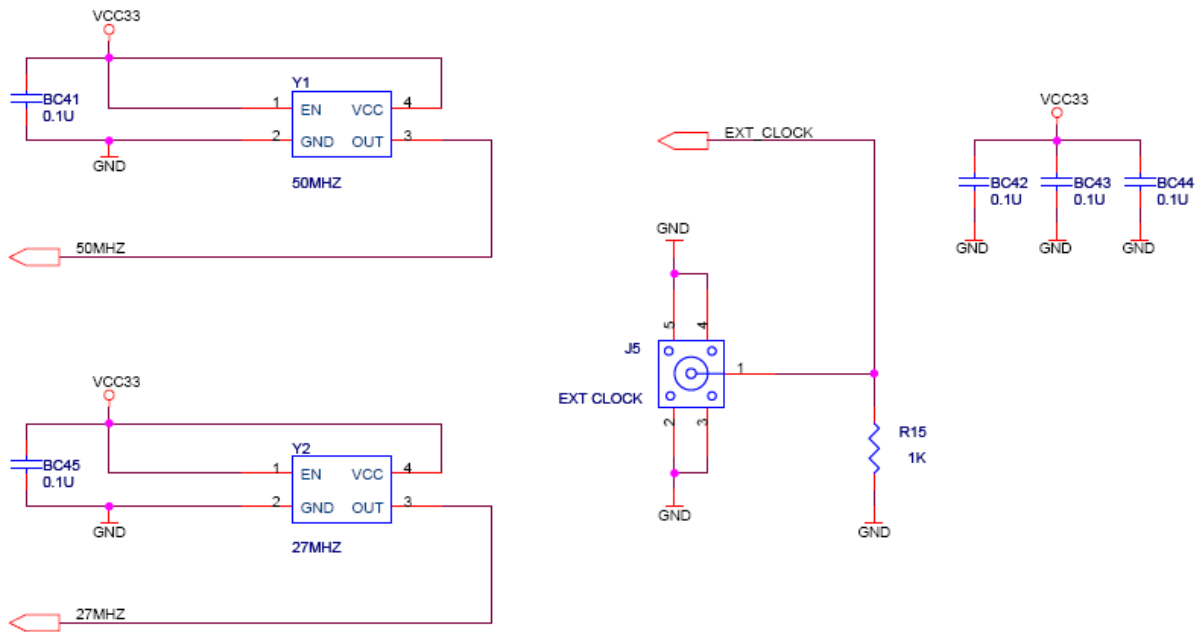


Figure 4.8. Schematic diagram of the clock circuit.

Signal Name	FPGA Pin No.	Description
CLOCK_27	PIN_D12, PIN_E12	27 MHz clock input
CLOCK_50	PIN_L1	50 MHz clock input
CLOCK_24	PIN_A12, PIN_B12	24 MHz clock input from USB Blaster
EXT_CLOCK	PIN_M21	External (SMA) clock input

Table 4.5. Pin assignments for the clock inputs.

4.5 Using the Expansion Header

The DE1 Board provides two 40-pin expansion headers. Each header connects directly to 36 pins on the Cyclone II FPGA, and also provides DC +5V (VCC5), DC +3.3V (VCC33), and two GND pins. Figure 4.10 shows the related schematics. Each pin on the expansion headers is connected to a resistor that provide protection from high and low voltages. The figure shows the protection circuitry for only four of the pins on each header, but this circuitry is included for all 72 data pins. Table 4.7 gives the pin assignments.

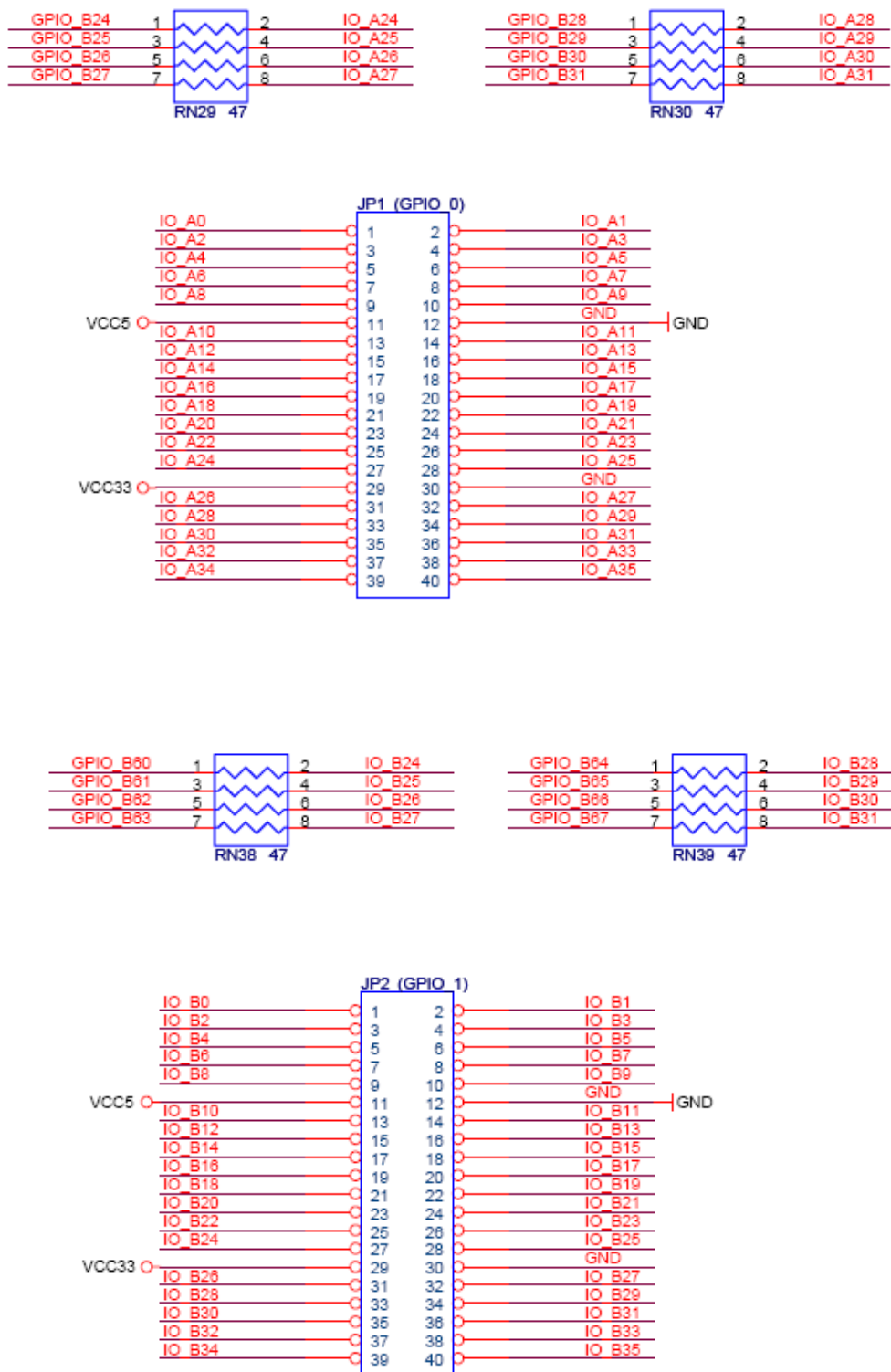


Figure 4.10. Schematic diagram of the expansion headers.

Signal Name	FPGA Pin No.	Description
GPIO_0[0]	PIN_A13	GPIO Connection 0[0]

GPIO_0[1]	PIN_B13	GPIO Connection 0[1]
GPIO_0[2]	PIN_A14	GPIO Connection 0[2]
GPIO_0[3]	PIN_B14	GPIO Connection 0[3]
GPIO_0[4]	PIN_A15	GPIO Connection 0[4]
GPIO_0[5]	PIN_B15	GPIO Connection 0[5]
GPIO_0[6]	PIN_A16	GPIO Connection 0[6]
GPIO_0[7]	PIN_B16	GPIO Connection 0[7]
GPIO_0[8]	PIN_A17	GPIO Connection 0[8]
GPIO_0[9]	PIN_B17	GPIO Connection 0[9]
GPIO_0[10]	PIN_A18	GPIO Connection 0[10]
GPIO_0[11]	PIN_B18	GPIO Connection 0[11]
GPIO_0[12]	PIN_A19	GPIO Connection 0[12]
GPIO_0[13]	PIN_B19	GPIO Connection 0[13]
GPIO_0[14]	PIN_A20	GPIO Connection 0[14]
GPIO_0[15]	PIN_B20	GPIO Connection 0[15]
GPIO_0[16]	PIN_C21	GPIO Connection 0[16]
GPIO_0[17]	PIN_C22	GPIO Connection 0[17]
GPIO_0[18]	PIN_D21	GPIO Connection 0[18]
GPIO_0[19]	PIN_D22	GPIO Connection 0[19]
GPIO_0[20]	PIN_E21	GPIO Connection 0[20]
GPIO_0[21]	PIN_E22	GPIO Connection 0[21]
GPIO_0[22]	PIN_F21	GPIO Connection 0[22]
GPIO_0[23]	PIN_F22	GPIO Connection 0[23]
GPIO_0[24]	PIN_G21	GPIO Connection 0[24]
GPIO_0[25]	PIN_G22	GPIO Connection 0[25]
GPIO_0[26]	PIN_J21	GPIO Connection 0[26]
GPIO_0[27]	PIN_J22	GPIO Connection 0[27]
GPIO_0[28]	PIN_K21	GPIO Connection 0[28]
GPIO_0[29]	PIN_K22	GPIO Connection 0[29]
GPIO_0[30]	PIN_J19	GPIO Connection 0[30]
GPIO_0[31]	PIN_J20	GPIO Connection 0[31]
GPIO_0[32]	PIN_J18	GPIO Connection 0[32]
GPIO_0[33]	PIN_K20	GPIO Connection 0[33]
GPIO_0[34]	PIN_L19	GPIO Connection 0[34]
GPIO_0[35]	PIN_L18	GPIO Connection 0[35]
GPIO_1[0]	PIN_H12	GPIO Connection 1[0]
GPIO_1[1]	PIN_H13	GPIO Connection 1[1]

GPIO_1[2]	PIN_H14	GPIO Connection 1[2]
GPIO_1[3]	PIN_G15	GPIO Connection 1[3]
GPIO_1[4]	PIN_E14	GPIO Connection 1[4]
GPIO_1[5]	PIN_E15	GPIO Connection 1[5]
GPIO_1[6]	PIN_F15	GPIO Connection 1[6]
GPIO_1[7]	PIN_G16	GPIO Connection 1[7]
GPIO_1[8]	PIN_F12	GPIO Connection 1[8]
GPIO_1[9]	PIN_F13	GPIO Connection 1[9]
GPIO_1[10]	PIN_C14	GPIO Connection 1[10]
GPIO_1[11]	PIN_D14	GPIO Connection 1[11]
GPIO_1[12]	PIN_D15	GPIO Connection 1[12]
GPIO_1[13]	PIN_D16	GPIO Connection 1[13]
GPIO_1[14]	PIN_C17	GPIO Connection 1[14]
GPIO_1[15]	PIN_C18	GPIO Connection 1[15]
GPIO_1[16]	PIN_C19	GPIO Connection 1[16]
GPIO_1[17]	PIN_C20	GPIO Connection 1[17]
GPIO_1[18]	PIN_D19	GPIO Connection 1[18]
GPIO_1[19]	PIN_D20	GPIO Connection 1[19]
GPIO_1[20]	PIN_E20	GPIO Connection 1[20]
GPIO_1[21]	PIN_F20	GPIO Connection 1[21]
GPIO_1[22]	PIN_E19	GPIO Connection 1[22]
GPIO_1[23]	PIN_E18	GPIO Connection 1[23]
GPIO_1[24]	PIN_G20	GPIO Connection 1[24]
GPIO_1[25]	PIN_G18	GPIO Connection 1[25]
GPIO_1[26]	PIN_G17	GPIO Connection 1[26]
GPIO_1[27]	PIN_H17	GPIO Connection 1[27]
GPIO_1[28]	PIN_J15	GPIO Connection 1[28]
GPIO_1[29]	PIN_H18	GPIO Connection 1[29]
GPIO_1[30]	PIN_N22	GPIO Connection 1[30]
GPIO_1[31]	PIN_N21	GPIO Connection 1[31]
GPIO_1[32]	PIN_P15	GPIO Connection 1[32]
GPIO_1[33]	PIN_N15	GPIO Connection 1[33]
GPIO_1[34]	PIN_P17	GPIO Connection 1[34]
GPIO_1[35]	PIN_P18	GPIO Connection 1[35]

Table 4.7. Pin assignments for the expansion headers.

4.6 Using VGA

The DE1 board includes a 16-pin D-SUB connector for VGA output. The VGA synchronization signals are provided directly from the Cyclone II FPGA, and a 4-bit DAC using resistor network is used to produce the analog data signals (red, green, and blue). The associated schematic is given in Figure 4.11 and can support standard VGA resolution (640x480 pixels, at 25 MHz).

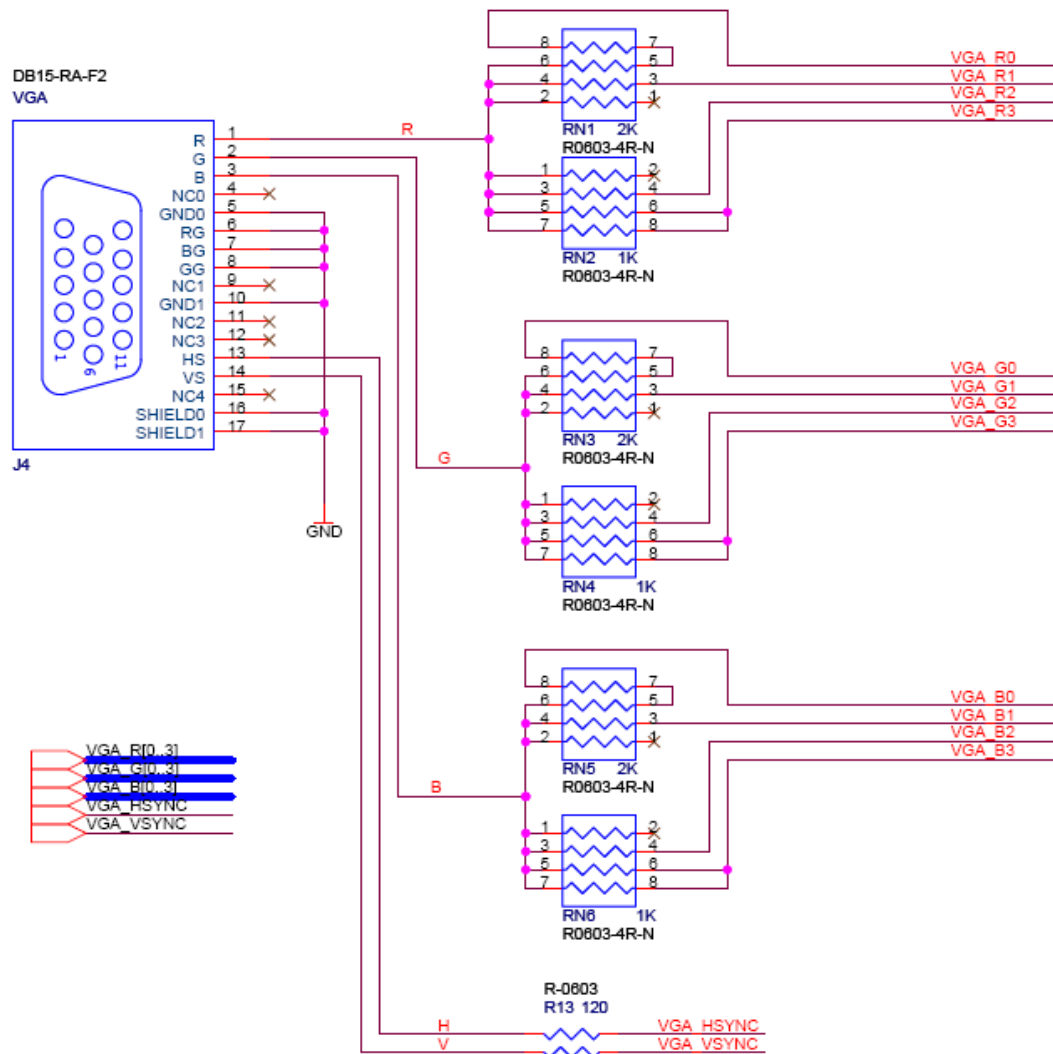


Figure 4.11. VGA circuit schematic.

The timing specification for VGA synchronization and RGB (red, green, blue) data can be found on various educational web sites (for example, search for “VGA signal timing”). Figure 4.12 illustrates the basic timing requirements for each row (horizontal) that is displayed on a VGA monitor. An active-low pulse of specific duration (time a in the figure) is applied to the horizontal synchronization ($hsync$) input of the monitor, which signifies the end of one row of data and the start of the next. The data (RGB) inputs on the monitor must be off (driven to 0 V) for a time period called the *back porch* (b) after the $hsync$ pulse occurs, which is followed by the display interval (c).

During the data display interval the RGB data drives each pixel in turn across the row being displayed. Finally, there is a time period called the *front porch* (*d*) where the RGB signals must again be off before the next *hsync* pulse can occur. The timing of the vertical synchronization (*vsync*) is the same as shown in Figure 4.12, except that a *vsync* pulse signifies the end of one frame and the start of the next, and the data refers to the set of rows in the frame (horizontal timing). Figures 4.13 and 4.14 show, for different resolutions, the durations of time periods *a*, *b*, *c*, and *d* for both horizontal and vertical timing.

The pin assignments between the Cyclone II FPGA and the VGA connector are listed in Table 4.8. An example of code that drives a VGA display is described in Sections 5.2 and 5.3.

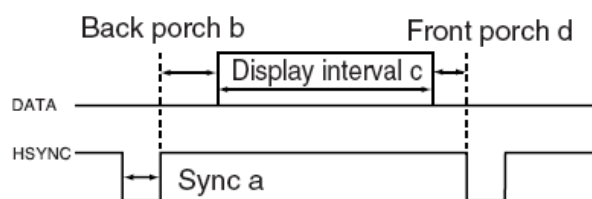


Figure 4.12. VGA horizontal timing specification.

VGA mode		Horizontal Timing Spec				
Configuration	Resolution(HxV)	a(us)	b(us)	c(us)	d(us)	Pixel clock(Mhz)
VGA(60Hz)	640x480	3.8	1.9	25.4	0.6	25 (640/c)

Figure 4.13. VGA horizontal timing specification.

VGA mode		Vertical Timing Spec			
Configuration	Resolution (HxV)	a(lines)	b(lines)	c(lines)	d(lines)
VGA(60Hz)	640x480	2	33	480	10

Figure 4.14. VGA vertical timing specification.

Signal Name	FPGA Pin No.	Description
VGA_R[0]	PIN_D9	VGA Red[0]
VGA_R[1]	PIN_C9	VGA Red[1]
VGA_R[2]	PIN_A7	VGA Red[2]
VGA_R[3]	PIN_B7	VGA Red[3]
VGA_G[0]	PIN_B8	VGA Green[0]
VGA_G[1]	PIN_C10	VGA Green[1]
VGA_G[2]	PIN_B9	VGA Green[2]
VGA_G[3]	PIN_A8	VGA Green[3]

VGA_B[0]	PIN_A9	VGA Blue[0]
VGA_B[1]	PIN_D11	VGA Blue[1]
VGA_B[2]	PIN_A10	VGA Blue[2]
VGA_B[3]	PIN_B10	VGA Blue[3]
VGA_HS	PIN_A11	VGA H_SYNC
VGA_VS	PIN_B11	VGA V_SYNC

Table 4.8. ADV7123 pin assignments.

4.7 Using the 24-bit Audio CODEC

The DE1 board provides high-quality 24-bit audio via the Wolfson WM8731 audio CODEC (enCOder/DECOder). This chip supports microphone-in, line-in, and line-out ports, with a sample rate adjustable from 8 kHz to 96 kHz. The WM8731 is controlled by a serial I2C bus interface, which is connected to pins on the Cyclone II FPGA. A schematic diagram of the audio circuitry is shown in Figure 4.15, and the FPGA pin assignments are listed in Table 4.9. Detailed information for using the WM8731 codec is available in its datasheet, which can be found on the manufacturer's web site, and from the *Datasheet* folder on the **DE1 System CD-ROM**.

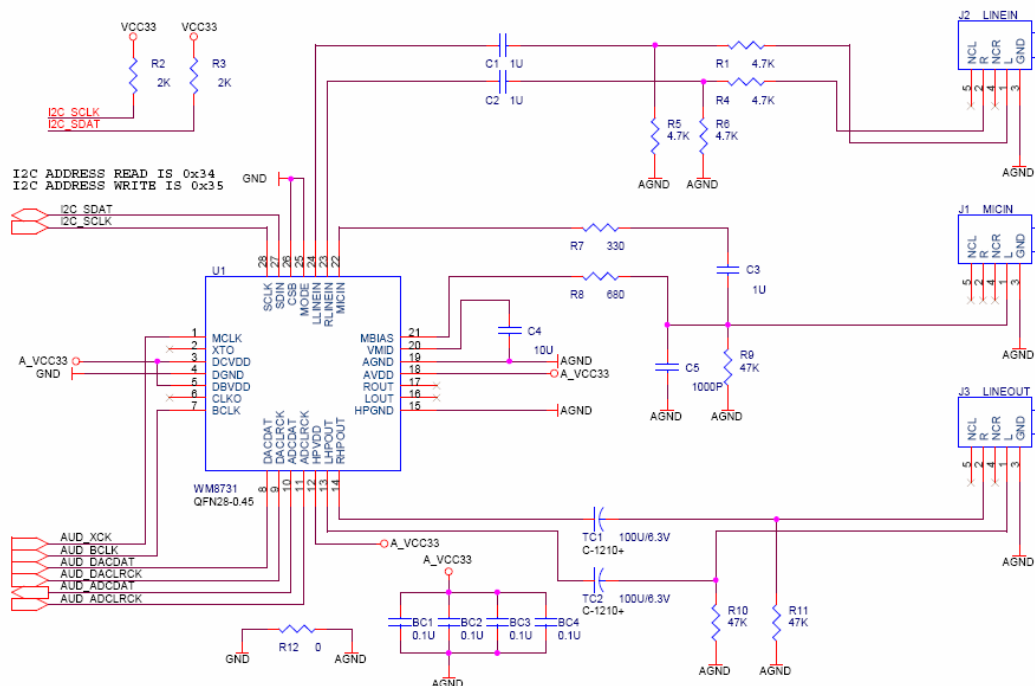


Figure 4.15. Audio CODEC schematic.

Signal Name	FPGA Pin No.	Description
AUD_ADCLRCK	PIN_A6	Audio CODEC ADC LR Clock
AUD_ADCDAT	PIN_B6	Audio CODEC ADC Data
AUD_DACLK	PIN_A5	Audio CODEC DAC LR Clock
AUD_DACDAT	PIN_B5	Audio CODEC DAC Data
AUD_XCK	PIN_B4	Audio CODEC Chip Clock
AUD_BCLK	PIN_A4	Audio CODEC Bit-Stream Clock
I2C_SCLK	PIN_A3	I2C Data
I2C_SDAT	PIN_B3	I2C Clock

Table 4.9. Audio CODEC pin assignments.

4.8 RS-232 Serial Port

The DE1 board uses the MAX232 transceiver chip and a 9-pin D-SUB connector for RS-232 communications. For detailed information on how to use the transceiver refer to the datasheet, which is available on the manufacturer's web site, and from the *Datasheet* folder on the **DE1 System CD-ROM**. Figure 4.16 shows the related schematics, and Table 4.10 lists the Cyclone II FPGA pin assignments.

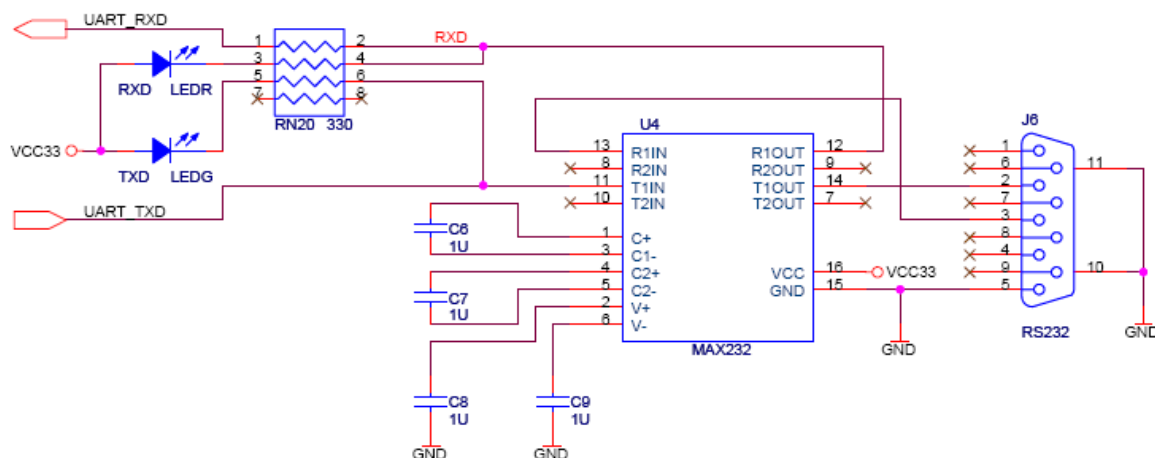


Figure 4.16. MAX232 (RS-232) chip schematic.

Signal Name	FPGA Pin No.	Description
UART_RXD	PIN_F14	UART Receiver
UART_TXD	PIN_G12	UART Transmitter

Table 4.10. RS-232 pin assignments.

4.9 PS/2 Serial Port

The DE1 board includes a standard PS/2 interface and a connector for a PS/2 keyboard or mouse. Figure 4.17 shows the schematic of the PS/2 circuit. Instructions for using a PS/2 mouse or keyboard can be found by performing an appropriate search on various educational web sites. The pin assignments for the associated interface are shown in Table 4.11.

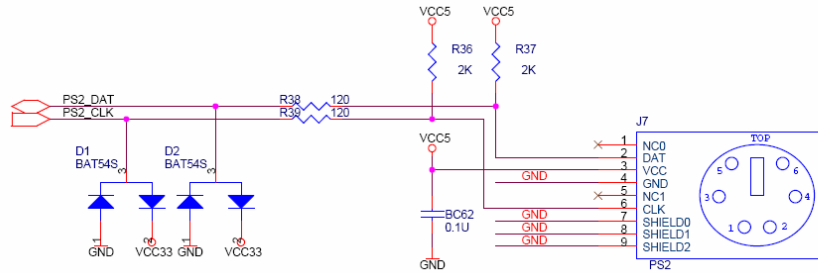


Figure 4.17. PS/2 schematic.

Signal Name	FPGA Pin No.	Description
PS2_CLK	PIN_H15	PS/2 Clock
PS2_DAT	PIN_J14	PS/2 Data

Table 4.11. PS/2 pin assignments.

4.10 Using SDRAM/SRAM/Flash

The DE1 board provides an 8-Mbyte SDRAM, 512-Kbyte SRAM, and 4-Mbyte (1-Mbyte on some boards) Flash memory. Figures 4.23, 4.24, and 4.25 show the schematics of the memory chips. The pin assignments for each device are listed in Tables 4.16, 4.17, and 4.18. The datasheets for the memory chips are provided in the *Datasheet* folder on the **DE1 System CD-ROM**.

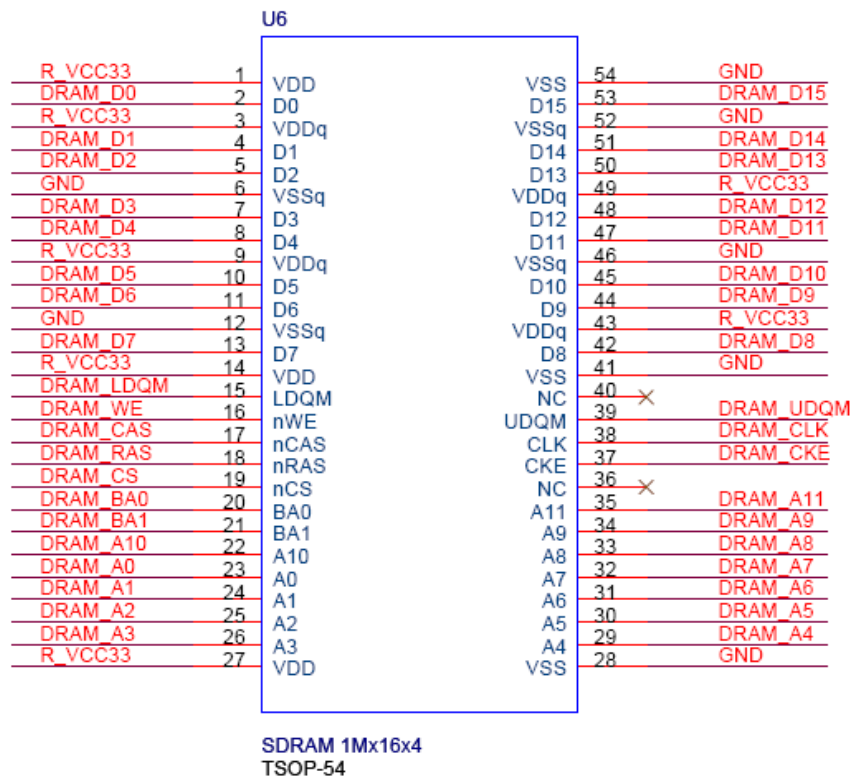


Figure 4.23. SDRAM schematic.

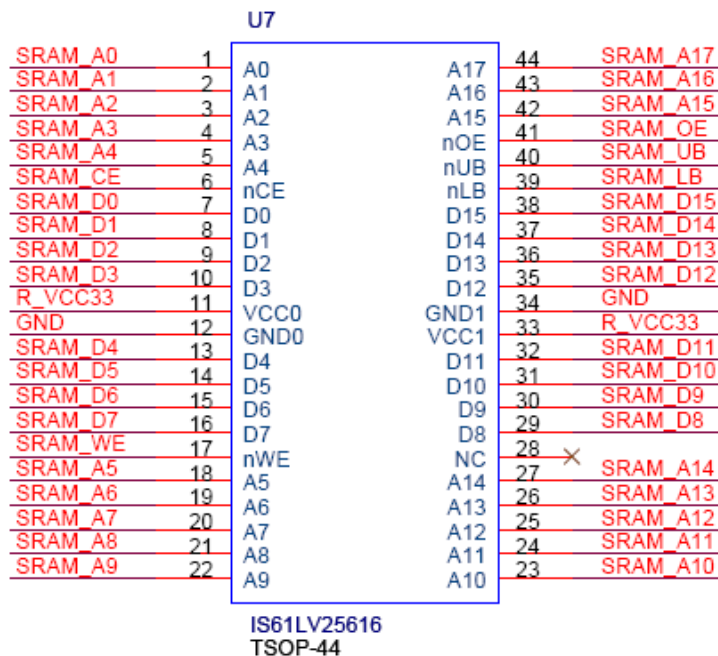


Figure 4.24. SRAM schematic.

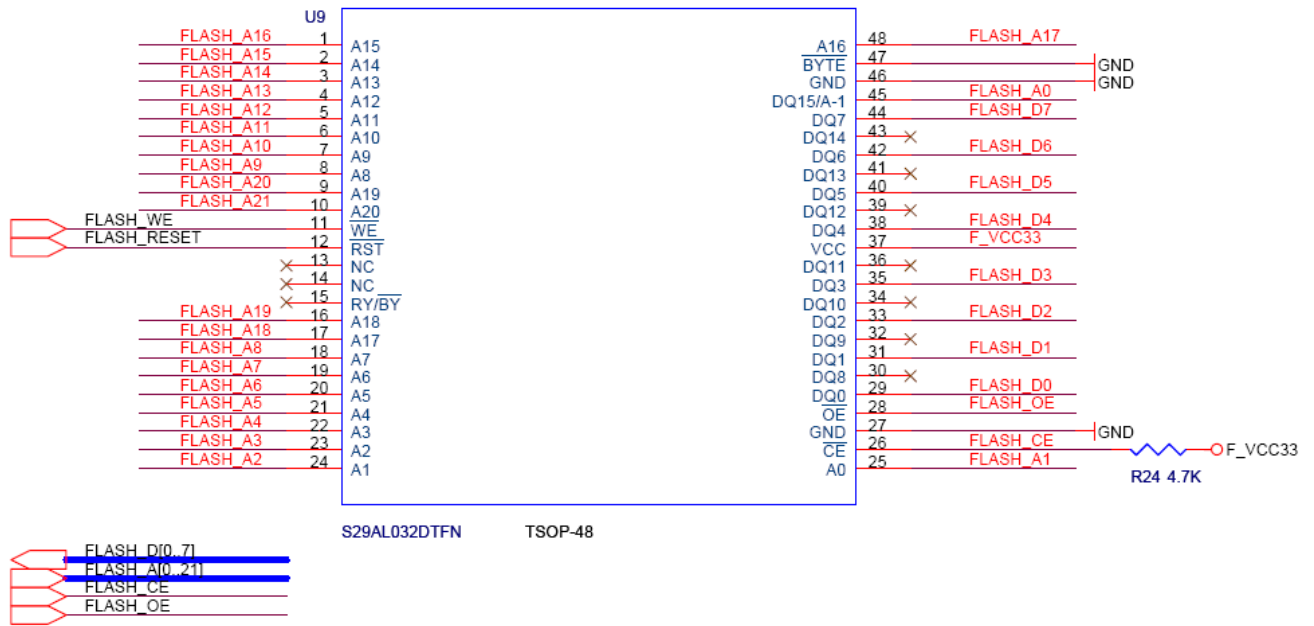


Figure 4.25. Flash schematic.

Signal Name	FPGA Pin No.	Description
DRAM_ADDR[0]	PIN_W4	SDRAM Address[0]
DRAM_ADDR[1]	PIN_W5	SDRAM Address[1]
DRAM_ADDR[2]	PIN_Y3	SDRAM Address[2]
DRAM_ADDR[3]	PIN_Y4	SDRAM Address[3]
DRAM_ADDR[4]	PIN_R6	SDRAM Address[4]
DRAM_ADDR[5]	PIN_R5	SDRAM Address[5]
DRAM_ADDR[6]	PIN_P6	SDRAM Address[6]
DRAM_ADDR[7]	PIN_P5	SDRAM Address[7]
DRAM_ADDR[8]	PIN_P3	SDRAM Address[8]
DRAM_ADDR[9]	PIN_N4	SDRAM Address[9]
DRAM_ADDR[10]	PIN_W3	SDRAM Address[10]
DRAM_ADDR[11]	PIN_N6	SDRAM Address[11]
DRAM_DQ[0]	PIN_U1	SDRAM Data[0]
DRAM_DQ[1]	PIN_U2	SDRAM Data[1]
DRAM_DQ[2]	PIN_V1	SDRAM Data[2]
DRAM_DQ[3]	PIN_V2	SDRAM Data[3]
DRAM_DQ[4]	PIN_W1	SDRAM Data[4]
DRAM_DQ[5]	PIN_W2	SDRAM Data[5]
DRAM_DQ[6]	PIN_Y1	SDRAM Data[6]

DRAM_DQ[7]	PIN_Y2	SDRAM Data[7]
DRAM_DQ[8]	PIN_N1	SDRAM Data[8]
DRAM_DQ[9]	PIN_N2	SDRAM Data[9]
DRAM_DQ[10]	PIN_P1	SDRAM Data[10]
DRAM_DQ[11]	PIN_P2	SDRAM Data[11]
DRAM_DQ[12]	PIN_R1	SDRAM Data[12]
DRAM_DQ[13]	PIN_R2	SDRAM Data[13]
DRAM_DQ[14]	PIN_T1	SDRAM Data[14]
DRAM_DQ[15]	PIN_T2	SDRAM Data[15]
DRAM_BA_0	PIN_U3	SDRAM Bank Address[0]
DRAM_BA_1	PIN_V4	SDRAM Bank Address[1]
DRAM_LDQM	PIN_R7	SDRAM Low-byte Data Mask
DRAM_UDQM	PIN_M5	SDRAM High-byte Data Mask
DRAM_RAS_N	PIN_T5	SDRAM Row Address Strobe
DRAM_CAS_N	PIN_T3	SDRAM Column Address Strobe
DRAM_CKE	PIN_N3	SDRAM Clock Enable
DRAM_CLK	PIN_U4	SDRAM Clock
DRAM_WE_N	PIN_R8	SDRAM Write Enable
DRAM_CS_N	PIN_T6	SDRAM Chip Select

Table 4.16. SDRAM pin assignments.

Signal Name	FPGA Pin No.	Description
SRAM_ADDR[0]	PIN_AA3	SRAM Address[0]
SRAM_ADDR[1]	PIN_AB3	SRAM Address[1]
SRAM_ADDR[2]	PIN_AA4	SRAM Address[2]
SRAM_ADDR[3]	PIN_AB4	SRAM Address[3]
SRAM_ADDR[4]	PIN_AA5	SRAM Address[4]
SRAM_ADDR[5]	PIN_AB10	SRAM Address[5]
SRAM_ADDR[6]	PIN_AA11	SRAM Address[6]
SRAM_ADDR[7]	PIN_AB11	SRAM Address[7]
SRAM_ADDR[8]	PIN_V11	SRAM Address[8]
SRAM_ADDR[9]	PIN_W11	SRAM Address[9]
SRAM_ADDR[10]	PIN_R11	SRAM Address[10]
SRAM_ADDR[11]	PIN_T11	SRAM Address[11]
SRAM_ADDR[12]	PIN_Y10	SRAM Address[12]
SRAM_ADDR[13]	PIN_U10	SRAM Address[13]

SRAM_ADDR[14]	PIN_R10	SRAM Address[14]
SRAM_ADDR[15]	PIN_T7	SRAM Address[15]
SRAM_ADDR[16]	PIN_Y6	SRAM Address[16]
SRAM_ADDR[17]	PIN_Y5	SRAM Address[17]
SRAM_DQ[0]	PIN_AA6	SRAM Data[0]
SRAM_DQ[1]	PIN_AB6	SRAM Data[1]
SRAM_DQ[2]	PIN_AA7	SRAM Data[2]
SRAM_DQ[3]	PIN_AB7	SRAM Data[3]
SRAM_DQ[4]	PIN_AA8	SRAM Data[4]
SRAM_DQ[5]	PIN_AB8	SRAM Data[5]
SRAM_DQ[6]	PIN_AA9	SRAM Data[6]
SRAM_DQ[7]	PIN_AB9	SRAM Data[7]
SRAM_DQ[8]	PIN_Y9	SRAM Data[8]
SRAM_DQ[9]	PIN_W9	SRAM Data[9]
SRAM_DQ[10]	PIN_V9	SRAM Data[10]
SRAM_DQ[11]	PIN_U9	SRAM Data[11]
SRAM_DQ[12]	PIN_R9	SRAM Data[12]
SRAM_DQ[13]	PIN_W8	SRAM Data[13]
SRAM_DQ[14]	PIN_V8	SRAM Data[14]
SRAM_DQ[15]	PIN_U8	SRAM Data[15]
SRAM_WE_N	PIN_AA10	SRAM Write Enable
SRAM_OE_N	PIN_T8	SRAM Output Enable
SRAM_UB_N	PIN_W7	SRAM High-byte Data Mask
SRAM_LB_N	PIN_Y7	SRAM Low-byte Data Mask
SRAM_CE_N	PIN_AB5	SRAM Chip Enable

Table 4.17. SRAM pin assignments.

Signal Name	FPGA Pin No.	Description
FL_ADDR[0]	PIN_AB20	FLASH Address[0]
FL_ADDR[1]	PIN_AA14	FLASH Address[1]
FL_ADDR[2]	PIN_Y16	FLASH Address[2]
FL_ADDR[3]	PIN_R15	FLASH Address[3]
FL_ADDR[4]	PIN_T15	FLASH Address[4]
FL_ADDR[5]	PIN_U15	FLASH Address[5]
FL_ADDR[6]	PIN_V15	FLASH Address[6]
FL_ADDR[7]	PIN_W15	FLASH Address[7]

FL_ADDR[8]	PIN_R14	FLASH Address[8]
FL_ADDR[9]	PIN_Y13	FLASH Address[9]
FL_ADDR[10]	PIN_R12	FLASH Address[10]
FL_ADDR[11]	PIN_T12	FLASH Address[11]
FL_ADDR[12]	PIN_AB14	FLASH Address[12]
FL_ADDR[13]	PIN_AA13	FLASH Address[13]
FL_ADDR[14]	PIN_AB13	FLASH Address[14]
FL_ADDR[15]	PIN_AA12	FLASH Address[15]
FL_ADDR[16]	PIN_AB12	FLASH Address[16]
FL_ADDR[17]	PIN_AA20	FLASH Address[17]
FL_ADDR[18]	PIN_U14	FLASH Address[18]
FL_ADDR[19]	PIN_V14	FLASH Address[19]
FL_ADDR[20]	PIN_U13	FLASH Address[20]
FL_ADDR[21]	PIN_R13	FLASH Address[21]
FL_DQ[0]	PIN_AB16	FLASH Data[0]
FL_DQ[1]	PIN_AA16	FLASH Data[1]
FL_DQ[2]	PIN_AB17	FLASH Data[2]
FL_DQ[3]	PIN_AA17	FLASH Data[3]
FL_DQ[4]	PIN_AB18	FLASH Data[4]
FL_DQ[5]	PIN_AA18	FLASH Data[5]
FL_DQ[6]	PIN_AB19	FLASH Data[6]
FL_DQ[7]	PIN_AA19	FLASH Data[7]
FL_OE_N	PIN_AA15	FLASH Output Enable
FL_RST_N	PIN_W14	FLASH Reset
FL_WE_N	PIN_Y14	FLASH Write Enable

Table 4.18. Flash pin assignments.

Chapter 5

Examples of Advanced Demonstrations

This chapter provides a number of examples of advanced circuits implemented on the DE1 board. These circuits provide demonstrations of the major features on the board, such as its audio and video capabilities. For each demonstration the Cyclone II FPGA (or EPCS4 serial EEPROM) configuration file is provided, as well as the full source code in Verilog HDL code. All of the associated files can be found in the *DE1_demonstrations* folder from the **DE1 System CD-ROM**. For each of demonstrations described in the following sections, we give the name of the project directory for its files, which are subdirectories of the *DE1_demonstrations* folder.

Installing the Demonstrations

To install the demonstrations on your computer, perform the following

1. Copy the directory *DE1_demonstrations* into a local directory of your choice. It is important to ensure that the path to your local directory contains no spaces – otherwise, the Nios II software will not work.
2. In the directory *DE1_demonstrations*, go to the subdirectory *fixpaths*.
3. Run the *DE1_fixpaths.bat* batch file. In the dialog box that pops up, select the directory *DE1_demonstrations* in your local directory where you copied the files to. Click OK.
4. When *fixpaths* is finished, press any key to complete the process.

5.1 DE1 Factory Configuration

The DE1 board is shipped from the factory with a default configuration that demonstrates some of the basic features of the board. The setup required for this demonstration, and the locations of its files are shown below.

Demonstration Setup, File Locations, and Instructions

- Project directory: *DE1_Default*
- Bit stream used: *DE1_Default.sof* or *DE1_Default.pof*

- Power on the DE1 board, with the USB cable connected to the USB Blaster port. If necessary (that is, if the default factory configuration of the DE1 board is not currently stored in EPCS4 device), download the bit stream to the board by using either JTAG or AS programming
- You should now be able to observe that the 7-segment displays are displaying a sequence of characters, and the red and green LEDs are flashing.
- Optionally connect a VGA display to the VGA D-SUB connector. When connected, the VGA display should show a pattern of colors.
- Optionally connect a powered speaker to the stereo audio-out jack
- Place toggle switch SW9 in the UP position to hear a 1 kHz humming sound from the audio-out port. Alternatively, if switch SW9 is DOWN, the microphone-in port can be connected to a microphone to hear voice sounds, or the line-in port can be used to play audio from an appropriate sound source.

The Verilog source code for this demonstration is provided in the *DE1_Default* folder, which also includes the necessary files for the corresponding Quartus II project. The top-level Verilog file, called *DE1_Default.v*, can be used as a template for other projects, because it defines ports that correspond to all of the user-accessible pins on the Cyclone II FPGA.

5.2 Music Synthesizer Demonstration

This demonstration shows how to implement a Multi-tone Electronic Keyboard using DE1 board with a PS/2 Keyboard and a speaker. Figure 5.1 shows the setup of the demonstration.

PS/2 Keyboard is used as the piano keyboard for input. The Cyclone II FPGA on the DE1 board serves as the Music Synthesizer SOC to generate music and tones. The VGA connected to the DE1 board is used to show which key is pressed during the playing of the music.

Figure 5.2 shows the block diagram of the design of the Music Synthesizer. There are four major blocks in the circuit: DEMO_SOUND, PS2_KEYBOARD, STAFF, and TONE_GENERATOR. The DEMO_SOUND block stores a demo sound for user to play; PS2_KEYBOARD handles the users' input from PS/2 keyboard; The STAFF block draws the corresponding keyboard diagram on VGA monitor when key(s) are pressed. The TONE_GENERATOR is the core of music synthesizer SOC.

User can switch the music source either from PS2_KEYBOARD or the DEMO_SOUND block using SW9. To repeat the demo sound, users can press KEY1.

The TONE_GENERATOR has two tones: (1) String. (2) Brass, which can be controlled by SW0.

The audio codec used on the DE1 board has two channels, which can be turned ON/OFF using SW1 and SW2.

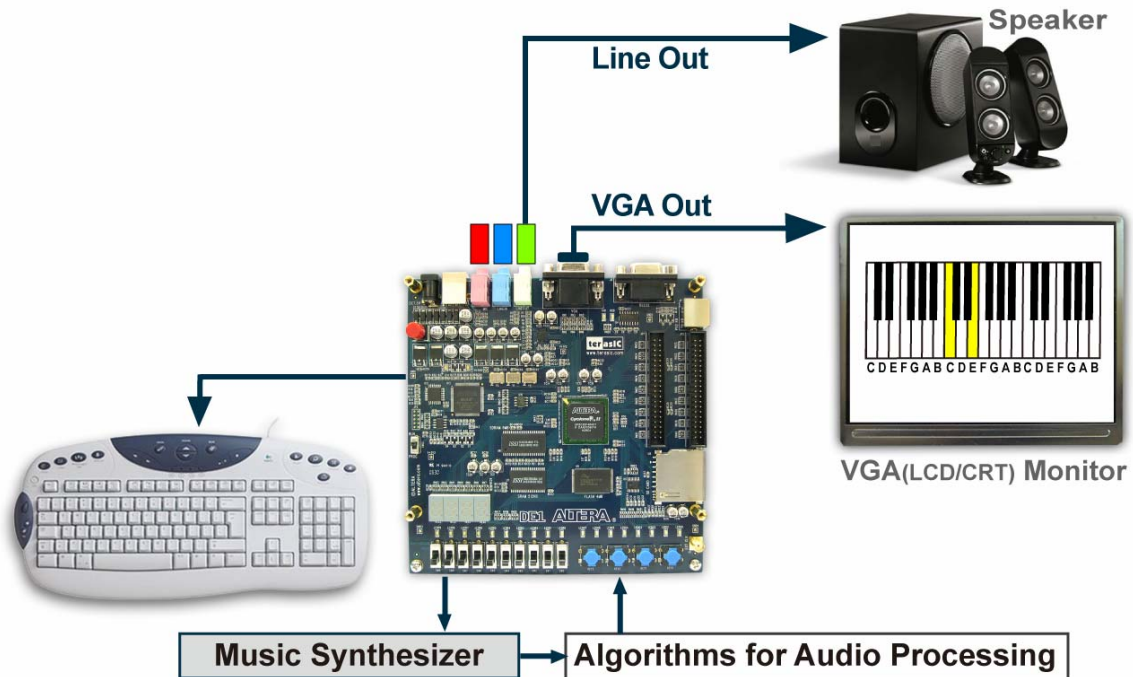


Figure 5.1. The Setup of the Music Synthesizer Demonstration.

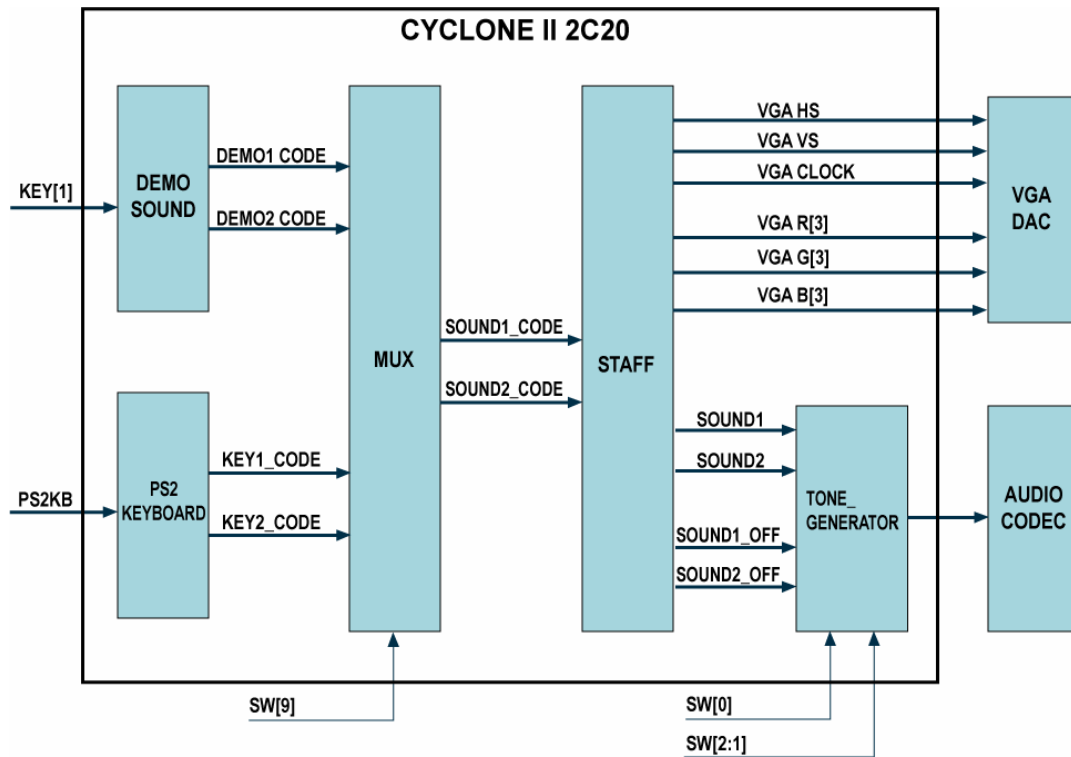


Figure 5.2. Block diagram of the Music Synthesizer design

Demonstration Setup, File Locations, and Instructions

- Project directory: *DE1_Synthesizer*
- Bit stream used: *DE1_Synthesizer.sof* or *DE1_Synthesizer.pof*
- Connect a PS/2 Keyboard to the DE1 board.
- Connect the VGA output of the DE1 board to a VGA monitor (both LCD and CRT type of monitors should work)
- Connect the Lineout of the DE1 board to a speaker.
- Load the bit stream into FPGA.
- Make sure all the switches (SW[9:0]) are set to 0 (Down Position)
- Press KEY1 on the DE1 board to start the music demo
- Press KEY0 on the DE1 board to reset the circuit

Figure 5.3 illustrates the usage of the switches, pushbuttons (KEYs), PS/2 Keyboard.

- Switches and Pushbuttons

Signal Name	Description
KEY[0]	Reset Circuit
KEY[1]	Repeat the Demo Music
SW[0]	OFF: BRASS, ON: STRING
SW[9]	OFF: DEMO, ON: PS2 KEYBOARD
SW[1]	Channel-1 ON / OFF
SW[2]	Channel-2 ON / OFF

- PS/2 Keyboard

Signal Name	Description
Q	-#4
A	-5
W	-#5
S	-6
E	-#6
D	-7
F	1
T	#1
G	2
Y	#2
H	3
J	4

I	#4
K	5
O	#5
L	6
P	#6
:	7
"	+1

Figure 5.3. Usage of the Key

5.3 A Karaoke Machine

This demonstration uses the microphone-in, line-in, and line-out ports on the DE1 board to create a Karaoke Machine application. The Wolfson WM8731 audio CODEC is configured in the master mode, where the audio CODEC generates AD/DA serial bit clock (BCK) and the left/right channel clock (LRCK) automatically. As indicated in Figure 5.7, the I2C interface is used to configure the Audio CODEC. The sample rate and gain of the CODEC are set in this manner, and the data input from the line-in port is then mixed with the microphone-in port and the result is sent to the line-out port.

For this demonstration the sample rate is set to 48 kHz. Pressing the pushbutton *KEY0* reconfigures the gain of the audio CODEC via the I2C bus, cycling through one of the ten predefined gains (volume levels) provided by the device.

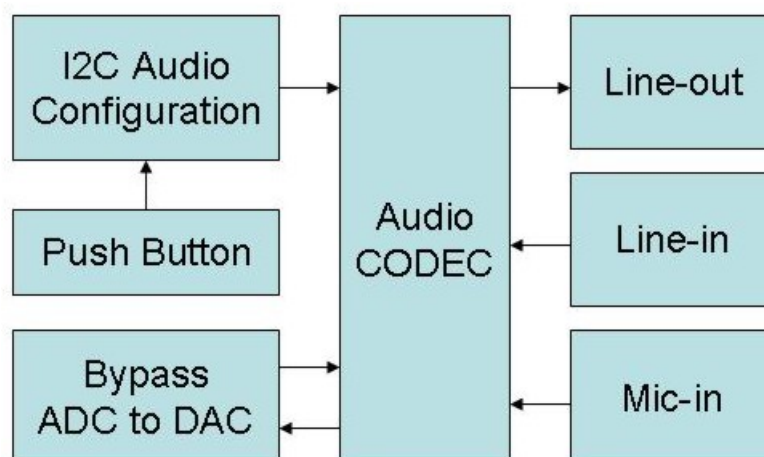


Figure 5.7. Block diagram of the Karaoke Machine demonstration.

Demonstration Setup, File Locations, and Instructions

- Project directory: *DE1_i2sound*
- Bit stream used: *DE1_i2sound.sof* or *DE1_i2sound.pof*
- Connect a microphone to the microphone-in port (pink color) on the DE1 board
- Connect the audio output of a music-player, such as an MP3 player or computer, to the line-in port (blue color) on the DE1 board
- Connect a headset/speaker to the line-out port (green color) on the DE1 board
- Load the bit stream into the FPGA
- You should be able to hear a mixture of the microphone sound and the sound from the music player
- Press *KEY0* to adjust the volume; it cycles between volume levels 0 to 9

Figure 5.8 illustrates the setup for this demonstration.

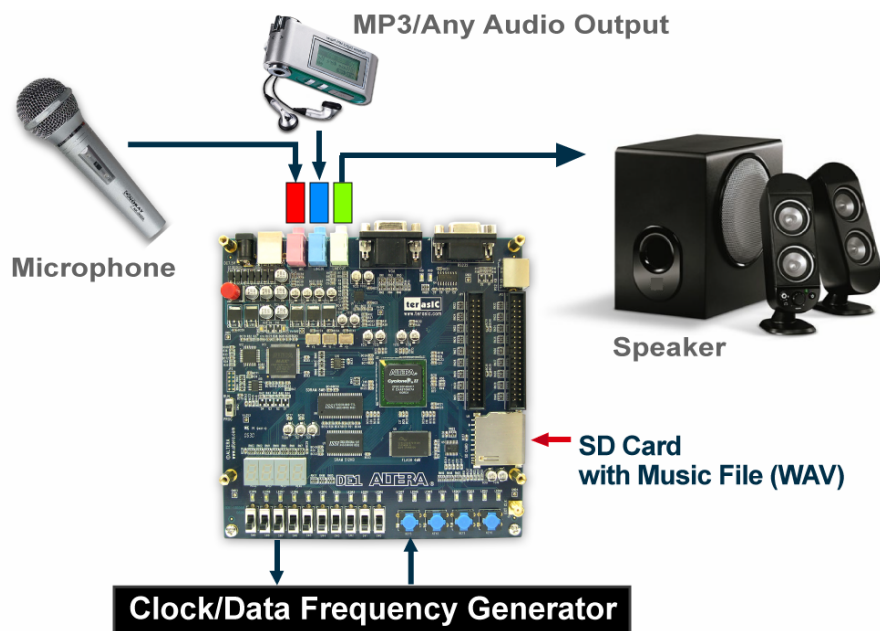


Figure 5.8. The setup for the Karaoke Machine.

5.4 SD Card Music Player

Many commercial media/audio players use a large external storage device, such as an SD card or CF card, to store music or video files. Such players may also include high-quality DAC devices so that good audio quality is produced. The DE1 board provides the hardware and software needed for SD card access and professional audio performance so that it is possible to design advanced multimedia products using the DE1 board.

In this demonstration we show how to implement an SD Card Music Player on the DE1 board, in which the music files are stored in an SD card and the board can play the music files via its CD-quality audio DAC circuits. We use the Nios II processor to read the music data stored in the SD Card and use the Wolfson WM8731 audio CODEC to play the music.

The audio CODEC is configured in the slave mode, where external circuitry must provide the ADC/DAC serial bit clock (*BCK*) and left/right channel clock (*LRCK*) to the audio CODEC. As shown in Figure 5.11, we provide an *Audio DAC Controller* to achieve the clock generation and the data flow control. The *Audio DAC Controller* is integrated into the Avalon bus architecture, so that the Nios II processor can control the application.

During operation the Nios II processor will check if the FIFO memory of the *Audio DAC Controller* becomes full. If the FIFO is not full, the processor will read a 512-byte sector and send the data to the FIFO of the *Audio DAC Controller* via the Avalon bus. The *Audio DAC Controller* uses a 48 kHz sample rate to send the data and clock signals to the audio CODEC. The design also mixes the data from microphone-in with line-in for the Karaoke-style effects.

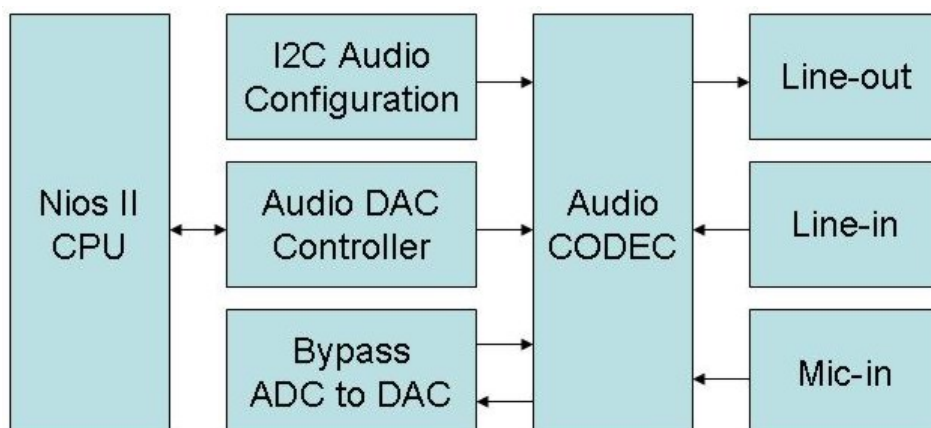


Figure 5.11. Block diagram of the SD music player demonstration.

Demonstration Setup, File Locations, and Instructions

- Project directory: *DE1_SD_Card_Audio*
- Bit stream used: *DE1_SD_Card_Audio.sof* (or *.pof*)
- Nios II Workspace: *DE1_SD_Card_Audio*
- Format your SD card into *FAT16* format
- To play a music file with this demonstration, the file must use the 48 KHz sample rate WAV format. Copy one or more such WAV files onto the FAT16-formatted SD Card. Due to a limitation in the software used for this demonstration, it is necessary to reformat the whole SD Card if any WAV file that has been copied onto the card needs to be later removed from the SD Card
- Load the bit stream into the FPGA
- Run the Nios II IDE under the workspace *DE1_SD_Card_Audio*
- Connect a headset or speaker to the DE1 board and you should be able to hear the music played from the SD Card

Figure 5.12 illustrates the setup for this demonstration.

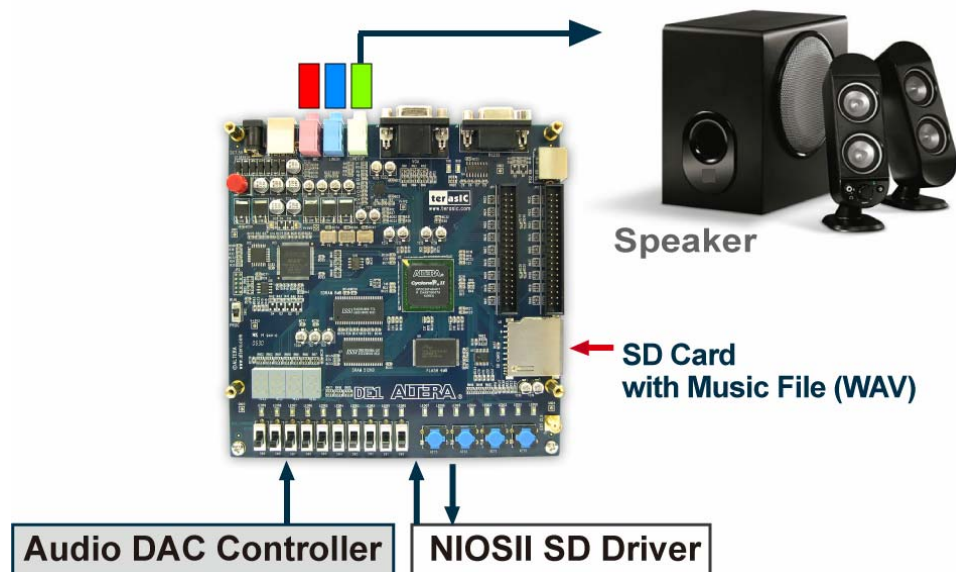


Figure 5.12. The setup for the SD music player demonstration.

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